

Merged research report for the Surreal project

# Definable Surreal Numbers and Omnific Integers

Ambient definability, reversible and bounded omnific coding,  
hereditary ordinal definability, a transcendence dichotomy,  
and the maximal initial core

Built from two manuscripts dated September 23, 2026

## Abstract

A surreal number cannot be called definable without specifying a language, a universe, and permitted parameters. We develop ambient set-theoretic notions, keep them apart from definability in the ordered field and its exponential expansion, and give an algebraic theory of the resulting numbers. For a fixed parameter set, the ambiently definable surreals form a real closed field closed under the Conway monomial map, the exponential and the omnific floor; their omnific integers form an integer part whose fraction field is exactly that field, with a denominator chosen from the same parameters. Over a transitive model the definable field is an elementary exponential subfield. Ordinal-definable surreals are exactly the surreals of HOD.

Two reversible omnific codes compress every surreal: a sign code with leading term  $\omega$ , zero constant term and coefficients in  $\{1, 2, 3\}$ , landing in the omnific interval  $\omega < z < \omega + \omega^{1/2}$ , and the monomial code  $x \mapsto \omega^{(1+x/\sqrt{1+x^2})/2}$ , landing between 0 and  $\omega$  and localizing to every infinitely wide omnific interval. Consequently ordinal definability of all omnific integers in one such interval is equivalent to  $V = \text{HOD}$ , and the parameter-free version in a model is equivalent to pointwise definability. If  $V \neq \text{HOD}$ , the first interval contains an explicit Ord-indexed algebraically independent family over the HOD surreals. If  $\rho$  is the first ordinal of a transitive model  $M$  not definable from  $A$ , then  $\rho$  is an epsilon number and the definable surreals of birthday below  $\rho$  form the largest initial subclass of the definable field; it is again an elementary exponential subfield and the fraction field of its own omnific integers. Explicit normal forms show in both directions that definability of a whole series differs from definability of its terms. We also treat complexity-bounded escape ordinals, homogeneous forcing that adds no reals, relativization, the surcomplex pairs, language dependence, and the finite quotients of the definable omnific ring.

This report merges two independently written manuscripts, prints shared results once with both sources named, and keeps every source's limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscript 04 (the base text) and manuscript 07, both pinned to repository commit [9a385d3](#); their titles are in [section 1.4](#). The first non-HOD birthday is printed by citation of the universes report. The provenance of every section is recorded in [appendix A](#); non-claims are collected in [appendix B](#). Finite checks accompany the report and test finite identities only.

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# 1 The question, the results, and how this report was assembled

## 1.1 The question

The familiar phrase “definable real number” conceals several choices. A positive root of a rational polynomial is definable in an ordered field; a computable real can instead be defined by a statement about its algorithm; and a set-theoretic definition can refer to ordinals, models, and forcing extensions. These are different notions, not alternative notations for one fixed countable field. The user-specified reference on definable reals is a useful starting point [18]; it is not used as a proof source. The foundational analysis and the pointwise-definability results used below come from Hamkins–Linetsky–Reitz [12].

For surreals the distinction becomes essential, because the sign sequence representing a number is a set while the totality of all surreals is a proper class internally. The pure ordered field has no parameter-free definable infinite element, whereas the first infinite ordinal  $\omega$  has a very short parameter-free definition in set theory. An omnific integer can simultaneously have elementary-looking normal-form coefficients and contain the information of an arbitrary set of ordinals. Our goal is to make these facts into a coherent theory rather than to announce an unspecified collection of “describable” numbers.

The central object is not a syntactic list of expressions and not a putative truth predicate inside the universe under discussion. Fix a model  $M$  of ZFC and an external collection  $A$  of allowed parameters. An element of  $M$  is *A-definable over M* when a standard first-order formula of set theory, using a finite tuple from  $A$ , uniquely specifies it in  $M$  (definition 3.1). We write  $D_A^M$  for the definable closure,  $K_A^M$  for its surreals and  $I_A^M$  for its omnific integers. When the universe  $V$  itself is meant, the same words are a metatheoretic scheme.

## 1.2 Main results

Let  $\mathbf{No}$  be represented by ordinal-length sign sequences, and let  $\mathbf{Oz}$  be the classical omnific-integer ring. Write

$$\mathcal{U} = \{z \in \mathbf{Oz} : \omega < z < \omega + \omega^{1/2}, \text{ ct}(z) = 0\}. \quad (1.1)$$

The exponentiation  $x \mapsto \omega^x$  throughout this report is the *Conway monomial map*, also written  $\Omega(x)$ ; it is not  $x \mapsto \exp(x \log \omega)$  (section 2.3).

**The definable field [04, 07].** For every parameter family the definable surreals  $K_A$  form a real closed field closed under  $\Omega$ ,  $\Omega^{-1}$  on monomials,  $\exp$ ,  $\log$  and the omnific floor;  $I_A = K_A \cap \mathbf{Oz}$  is an integer part and  $K_A = \text{Frac}(I_A)$  with a denominator chosen by one rule from the same parameters (theorem 4.1). Over a transitive model  $K_A^M \prec \mathbf{No}^M$  in the exponential language (theorem 4.4). The ordinal-definable surreals are the surreal field of HOD (theorem 5.1).

**Reversible compression [04, 07].** There is a parameter-free set-theoretically definable injection  $\mathbf{C} : \mathbf{No} \rightarrow \mathcal{U}$  with a parameter-free definable inverse on its image, preserving the definability degree for every parameter family (theorem 6.2, source 04). The monomial code  $\mathbf{Q}(x) = \omega^{\tau(x)}$ ,  $\tau(x) = \frac{1}{2}(1 + x/\sqrt{1+x^2})$ , is a parameter-free definable order embedding of  $\mathbf{No}$  into  $\mathbf{Oz} \cap (0, \omega)$  with the same property; it localizes to every infinitely wide omnific interval, and there is a definable class function  $\text{esc}$  on the ordinals with values in  $\mathbf{Oz} \cap (0, \omega)$  and unbounded birthdays (theorems 6.8 to 6.10, source 07). Neither code is asserted to be a ring homomorphism or to be definable in the pure ordered field.

**One interval tests the universe [04, 07].** Every element of  $\mathcal{U}$  is ordinal definable if and only if  $V = \text{HOD}$ , and the same holds for  $\mathbf{Oz} \cap (0, \omega)$  and for the monomials  $\omega^a$  with  $0 < a < 1$ . In

the external semantics of a model  $M \models \text{ZFC}$ , every element of  $\mathcal{U}^M$  is parameter-free definable in  $M$  if and only if  $M$  is pointwise definable (theorems 7.2 and 7.7). Thus a fixed leading asymptotic term, or a fixed numerical bound, does not bound set-theoretic information.

**A transcendence dichotomy [04].** Either  $\mathbf{No} = K_{\text{HOD}}$ , or  $\mathcal{U}$  contains an Ord-indexed algebraically independent family over  $K_{\text{HOD}}$  (theorem 8.3). The proof is explicit: starting with any  $t \in (0, 1) \setminus K_{\text{HOD}}$ , use

$$z_\alpha = \omega + \omega^{t\omega^{-(\alpha+1)}} \quad (\alpha \in \text{Ord}). \quad (1.2)$$

The obstruction to a polynomial relation is separation of normal-form supports into distinct additive cosets of  $K_{\text{HOD}}$ .

**The maximal initial core [07].** Let  $M$  be a transitive model. If some ordinal of  $M$  is not  $A$ -definable and  $\rho$  is the least such ordinal, taken externally, then  $\rho$  is an epsilon number, and

$$\text{Core}_A^M = K_A^M \cap (\mathbf{No}_{<\rho})^M$$

is the largest initial subclass of  $K_A^M$ . It is an elementary exponential subfield of  $\mathbf{No}^M$ , and it is the fraction field of its own omnific integers, with the birthday-controlled denominator  $\omega^{b(x)+1}$  (theorems 10.3 and 10.5).

**Two uniformity gaps [04, 07].** A fixed definable support and individually definable coefficients need not determine a definable coefficient assignment (theorems 9.1 and 9.3); conversely a uniformly definable omnific normal form can have a nondefinable term (proposition 9.5 and theorem 9.6). The correct invariant is the entire canonical normal-form graph (proposition 4.9).

**Further results.** The first birthday of a non-ordinal-definable surreal is the first subset disagreement between HOD and  $V$  (theorem 11.2, the case  $M = \text{HOD}$  of a theorem of the universes report); complexity-bounded definitions admit escape ordinals (theorem 12.3); no definable set contains its own definable hull (theorem 12.5); homogeneous forcing produces the dichotomy without adding reals (theorem 13.2); and the definable omnific ring has only the finite quotients  $\mathbb{Z}/n\mathbb{Z}$  (theorem 16.2).

### 1.3 What is prior work, and what is proposed here

The sign construction, normal forms, the monomial map, real closedness, the exponential, and the omnific integer part are classical [1, 2, 3, 4, 6]. Quantifier elimination, o-minimality, ordinal definability, HOD, partial truth predicates, and Mostowski collapse are standard tools [7, 8, 10, 12]. Van den Dries and Ehrlich supply the elementary exponential-field and epsilon-cutoff results [3, 4]; Bournez and Guilmant restate the corrected cutoff theorem [5]. We do not claim these tools, or their routine closure consequences, as new.

Hamkins proved in 2012 that the existence of a definable ordinal-indexed dense image in  $\mathbf{No}$ , and the density of the ordinal-definable surreals, are equivalent to  $V = \text{HOD}$  [13]. This density theorem is prior work; the bounded omnific tests below strengthen the form of the hypothesis, not the underlying idea (theorem 7.5). Source 04 did not cite it; the merge adds the credit.

Chen, Hamkins, and Yang have announced that the surreal ordered field expanded by birthday order is bi-interpretable with set theory [17]. The collection credits that announcement in its birthday-cutoff report (`hset:thm:chy`). In particular, the general idea of encoding well-founded membership graphs by surreal signs is *not* a new contribution of this report. No theorem below needs an unverified version of that announced bi-interpretation.

The candidate-original contributions are: the particular reversible omnific codes with fixed leading data or fixed numerical bounds, and their one-interval tests for  $V = \text{HOD}$  and for pointwise definability (04, 07); the explicit *proper-class transcendence dichotomy in the same interval* (04); the local coding, bounded birthday escape and bounded noncollection (07); the two explicit normal-form uniformity gaps (04, 07); and the maximal initial elementary exponential core with its birthday-controlled denominators (07). The birthday, complexity and forcing results are refinements and applications with less ambitious originality claims. Neither source’s literature search located these precise statements; neither search was exhaustive, and no theorem is described as settling a named open problem in the published literature. Mathematical correctness and bibliographic novelty are separate questions.

## 1.4 How this report was assembled

The report is built from two manuscripts, both dated 23 September 2026 and both pinned to repository commit 9a385d3957bdf9e3d9ea79f9a524751c90bd2c894 (35 commits before the placement commit a4dcb91); they are items 04 and 07 of batch 27 and keep those numbers here.

- **04:** *Definable Surreal Numbers and Omnific Integers: Ambient definability, reversible coding, hereditary ordinal definability, and a transcendence dichotomy.* It is the base text. It contributes the general set-model framework, the sign code and the two-term code, the one-interval tests (including the pointwise test for arbitrary set models), support-subfield amplification and the HOD dichotomy, localization, the termwise counterexample and the collector, the first non-HOD birthday, the  $\Sigma_n$  escape, forcing, relativization and the surcomplex companion.
- **07:** *Definable Surreal Numbers and Omnific Integers: Ambient truth, bounded coding, and the maximal initial core.* It works over an externally fixed transitive model throughout. It contributes the exponential elementarity of the definable field, the cut denominator rule, the valuation theorem, the monomial and local codes, the birthday escape and bounded noncollection, the bit code and the reverse gap, the maximal initial core and its denominators, Hamkins’s density theorem with the topology corollary, the language-dependence results, and the arithmetic of the definable omnific ring.

The source manuscripts are not shipped. Their code, recorded data and audits are, under the prefixes 04-hod-transcendence- and 07-initial-core-.

**Why one report, and why 04 is the base.** The two manuscripts have the same title and the same spine: the definable field and its integer part with same-parameter denominators, the HOD identification, bounded tests for  $V = \text{HOD}$  and for pointwise definability, the gap between whole normal forms and their terms, noncollection, and the definable closure of the pure field. Each has a large result the other lacks. On the shared theorems 04 has the weaker hypotheses: its pointwise test holds for arbitrary set models, including ill-founded ones, and its field theory is stated both for  $V$  as a scheme and for set models. 07’s stronger conclusions (exponential elementarity, the core) are proved under 07’s own transitive-model hypothesis, and they keep it here. The two manuscripts do not contradict each other or any report in the collection.

**Printed once.** Each of the following is printed once, crediting both sources: unique-output closure (lemma 3.2); the omnific floor (proposition 2.3); the definable closure of the pure field (theorem 3.4); the definable field, integer part and fractions (theorem 4.1, with both denominator rules); closure under whole definable cuts and sums (proposition 4.8); the HOD identification and the normal-form criterion (theorem 5.1 and lemma 5.2); the bounded  $V = \text{HOD}$  test

(theorem 7.2); the pointwise test (theorem 7.7); noncollection (theorem 12.5); the set-coding lemma (lemma 7.1); and the complexification (proposition 14.2).

**Kept as second routes.** 04's least-ordinal denominator and 07's cut denominator (theorem 4.1); 04's sign code and 07's bit code for the fixed-support phenomenon (theorems 9.1 and 9.3); 04's collector and 07's reverse gap for the converse phenomenon (proposition 9.5 and theorem 9.6); 04's two-term code and 07's monomial code (proposition 6.5 and theorem 6.8).

**Cited, not reprinted.** 04's first-birthday theorem is the case  $M = \text{HOD}$ ,  $N = V$  of `univ:prop:beta` in the universes report, with the same proof; it is printed by citation (theorem 11.2). 07's arithmetic section restricts results of the omnific Diophantine and quotient reports to definable fragments; the full-class antecedents are named at each statement (section 16).

**Added by the merge** , each tagged [merge] with a complete proof or an explicit citation:

- the fix of a gap in 04's proof that a proper  $K_{\text{HOD}}$  is not intrinsically definable (proposition 5.4);
- the remark on o-minimality of  $\mathbf{No}_{\text{exp}}^M$  used tacitly by 07 (remark 2.5);
- the extension of 07's bounded conditions of the pointwise test to arbitrary set models (theorem 7.7) and of 07's bounded noncollection to arbitrary parameter families (theorem 12.5);
- the answer from the collection to both sources' language questions (proposition 15.3), and the re-scoped question 17.1;
- the relation to `odg:q:size` (remark 16.6).

Appendix A.3 lists every stale statement corrected.

## 1.5 Conventions and renamed symbols

The two sources used several letters for different objects. The following convention is fixed before any use. Symbols of 04, the base, are kept where possible and 07's are renamed; the few local renamings in 04's text are listed as well.

- $D_A^M$  is the definable closure of  $A$  in  $(M, \in)$  (07's  $\text{Def}_M(A)$ );  $K_A^M = D_A^M \cap \mathbf{No}^M$  are the definable surreals (07's  $D_M(A)$ );  $I_A^M = K_A^M \cap \mathbf{Oz}^M$  (07's  $I_M(A)$ );  $O_A^M$  the definable ordinals;  $\rho_A^M$  the first omitted ordinal (07's  $\rho_M(A)$ ).
- $\text{Core}_A^M$  is 07's maximal initial core (07's  $K_M(A)$ ). The letter  $K$  never denotes the core.
- $\Pi$  is the purely infinite ideal (07's  $\mathcal{J}$ );  $\Pi_A = K_A \cap \Pi$  (07's  $\mathcal{J}_D$ );  $R_A = K_A \cap \mathbb{R}$  (07's  $R_D$ ).
- $\mathbf{C}$  is 04's sign code,  $\mathbf{D}$  its inverse;  $\mathbf{Q}$  and  $\mathbf{Q}_{a,b}$  are 07's monomial and local codes (07's  $\mathbf{C}$ ,  $\mathbf{C}_{a,b}$ ).
- $\mathbf{T}$  is 04's two-term code, built on the rational map  $h$ ; `esc` is 07's birthday escape (07's  $\mathbf{T}$ ). The rational map of 04's collector is renamed  $\zeta$ , and the width of 07's local code is  $\ell$  (07's  $h$ ).
- $e_\beta = \omega^{-(\beta+1)}$  (04) and  $\iota_\beta = 1/(\underline{\beta} + 2)$  (07's  $e_\beta$ ).
- $B_n$  is 04's set of  $\Sigma_n$ -definable surreals; 07's set of surreals of birthday at most  $\alpha$  is written  $\mathbf{No}_{\leq \alpha}$  (07's  $B_\alpha$ ), its subsets of ordinals  $X \subseteq \alpha$  (07's  $B$ ), and 04's ordinal code of a set is  $X_a$  (04's  $B$ ).

- 04’s decoder of sets from signs is  $\text{Set}$  (04’s  $E$ );  $E_\alpha(X)$  is 07’s bit code. In the set-coding lemma 04’s index  $\rho$  is written  $\varrho$ , and in the forcing lemma 04’s set  $A \subseteq \theta$  is written  $Y$ , since  $\rho$  and  $A$  are reserved.
- $\lambda$  is a normal-form length and  $\lambda_H$  the first non-HOD birthday; an epsilon number is written  $\epsilon$ , and an infinitesimal  $\varepsilon$  (07 used  $\lambda$  for epsilon numbers and for other ordinals).
- $\omega^x = \Omega(x)$  is the Conway map (07 wrote a bold  $\omega$  for the surreal  $\omega$ );  $\underline{\alpha}$  is the surreal ordinal  $\alpha$  when ordinal and surreal arithmetic might be confused.
- $\prec$  is elementary substructure;  $\prec_s$  is the proper-prefix (simplicity) relation.

Other reports use some of these letters differently: the automorphisms report writes  $\mathcal{U}_{\mathbf{Oz}}$  for a group, whereas  $\mathcal{U}$  here is an interval of  $\mathbf{Oz}$ ; the universes report writes  $\beta(M, N)$  for the first new birthday, which is  $\lambda_H$  when  $M = \text{HOD}$  and  $N = V$ .

## 1.6 Repository snapshot and dependency boundary

Both manuscripts inspected the repository at commit [9a385d3](#) through a GitHub connector: 04 read the catalogue and the introductions to the birthday-cutoff and omnific Diophantine reports [19, 20, 22]; 07 read the README, the catalogue and the detailed guides of the universes, birthday and omnific Diophantine reports [21]. The catalogue describes AI-assisted research drafts and separates written arguments from Lean coverage. This report follows that distinction. It imports no nonclassical Diophantine, quotient, automorphism, or bi-interpretation claim as a proof dependency; the relations to other reports stated below are credits and cross-references, checked at the placement commit. [Appendix A.4](#) records the limited audit scope of each source.

## 2 Foundations and canonical representations

### 2.1 Universes, models, and class notation

Our mathematical statements are made in ZFC, with proper-class notation understood schematically. Each individual surreal is a *set*. A class function used below is specified by one set-theoretic formula; all its values and all sums at which it is evaluated are set-sized. An Ord-indexed independent family means a definable class assignment whose every finite subfamily is algebraically independent. It does not mean that the entire family is a set or that its class sum exists.

When a literal satisfaction relation is needed, fix a set model  $M \models \text{ZFC}$  in an ambient metatheory, and interpret  $\mathbf{No}^M$ ,  $\mathbf{Oz}^M$ , and the relevant definitions internally. This conditional framework does not assert that ZFC proves the existence of such an  $M$ . The model-theoretic statements explicitly identified as external apply to arbitrary set models. For nonstandard models, their “ordinals,” “finite sequences,” and surreal codes are internal objects, not necessarily actual well-founded sequences. Where a proof uses a least *externally* chosen ordinal ([section 10](#)) or external o-minimality ([theorem 4.4](#)),  $M$  is assumed transitive, as in source 07; then its ordinals and finite sequences are genuine, and a surreal of  $M$  is literally an external sign sequence, although  $M$  may have only some of the subsets of a given ordinal.

### 2.2 Signs, simplicity, and cuts

A canonical surreal is a function

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

Its birthday is  $b(s) = \alpha$ . Comparison is by the first disagreement, with a missing sign ordered between  $-$  and  $+$ . The ordinal  $\alpha$  itself is represented by the all-plus sequence of length  $\alpha$ , written  $\underline{\alpha}$  when needed. The standard real embedding and the arithmetic operations are parameter-free definable in set theory [1, 2]. We write  $y \prec_s x$  when  $y$  is a proper prefix of  $x$ , and  $y \preceq_s x$  when equality is allowed.

A subclass  $F \subseteq \mathbf{No}$  is *initial* if it contains every prefix of each of its elements. This is not the same as being an initial segment of the numerical order. The notation  $\mathbf{No}_{<\lambda}$  always refers to birthdays, not numerical magnitude.

For sets  $L, R$  of surreal numbers with  $L < R$ , there is a unique simplest separator  $\{L \mid R\}$ . It is a prefix of every separator of the cut. Both the restriction that  $L, R$  be sets in the universe of construction and the word “simplest” are essential. These standard facts are part of the Conway–Gonshor simplicity theory; see also [9].

### 2.3 Normal forms and two different exponentials

Every nonzero surreal has a unique Conway normal form

$$x = \sum_{\xi < \lambda} r_\xi \omega^{g_\xi}, \quad r_\xi \in \mathbb{R} \setminus \{0\}, \quad g_\xi > g_\eta \quad (\xi < \eta < \lambda). \quad (2.1)$$

The length  $\lambda$  is an ordinal, and the support  $\text{supp}(x) = \{g_\xi : \xi < \lambda\}$  is reverse well-ordered and set-sized. Conversely, any such coefficient/exponent sequence has a surreal sum. Addition and multiplication are Hahn addition and convolution. We use  $\text{NF}(x)$  for the *whole indexed sequence*  $\xi \mapsto (g_\xi, r_\xi)$ , including its length, not merely the collection of its individual entries; for zero it is the empty function.

The uniqueness theorem supplies parameter-free set-theoretic operations  $x \mapsto \text{NF}(x)$  and evaluation of valid normal forms. It supplies no algorithm for arbitrary infinite codes and no assertion of topological convergence of finite partial sums. The factorisation paper [6, Sections 1–2] is a convenient modern reference for the normal-form convention and the distinction between series fields and omnific subrings.

We write  $\Omega(a) = \omega^a$  for the Conway omega map. It is an increasing isomorphism from the additive group of surreals to the multiplicative group of canonical monomials. Distinct exponents give distinct Archimedean scales: if  $a < b$ , then  $n\omega^a < \omega^b$  for every positive ordinary integer  $n$ . The inverse  $\Omega^{-1}$  is defined on monomials. Gonshor’s function  $\exp$  is a different function, extending ordinary real exponentiation with base  $e$ . Throughout,  $\mathcal{L}_{\exp}$  means the ordered-field language with this  $\exp$ , *not* with  $\Omega$ , and  $\mathcal{L}_{\text{or}}$  the ordered-field language. In general one must not replace  $\omega^a$  by  $\exp(a \log \omega)$ . All coding constructions below use the Conway omega map.

**Lemma 2.1** (Ordinal domination [04]). *Every set  $S \subseteq \mathbf{No}$  is strictly bounded in absolute value by a surreal ordinal. Moreover, the least ordinal  $\delta$  satisfying  $|s| < \delta$  for every  $s \in S$  is definable from  $S$ .*

*Proof.* A sign sequence of length  $\alpha$  lies between the all-minus and all-plus sequences of length  $\alpha$ , hence  $|s| \leq b(s)$ . Replacement gives the set of birthdays. An ordinal strictly exceeding all of them gives the required bound. The qualifying ordinals form a nonempty definable class, so it has a least member, uniquely specified from  $S$ .  $\square$

## 2.4 Omnific integers and the correct floor formula

**Definition 2.2.** An *omnific integer* is a surreal whose normal form has no negative exponents and whose coefficient at exponent 0 is an ordinary integer. Thus

$$\mathbf{Oz} = \Pi \oplus \mathbb{Z}, \quad \Pi = \{x \in \mathbf{No} : \text{supp}(x) \subseteq \mathbf{No}_{>0}\}. \quad (2.2)$$

The empty support is permitted, so  $0 \in \Pi$ . The map  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  takes the coefficient at exponent 0.

This is the standard ring, not the ring generated by the ordinals. For example,  $\sqrt{2}\omega$ ,  $\omega^{1/2}$  and  $\omega + 17$  are omnific, but  $\omega + \pi$ ,  $\omega + 1/2$  and  $\omega^{-1}$  are not. A positive omnific integer may be much smaller than  $\omega$  while exceeding every ordinary integer. The direct sum in (2.2) is additive, not a direct product of unital rings. Positive exponents stay positive under addition, so  $\Pi$  is an ideal and  $\text{ct}$  is a ring retraction.

**Proposition 2.3** (Integer part [04, 07]). *The ring  $\mathbf{Oz}$  is discretely ordered with least positive element 1. For every  $x \in \mathbf{No}$  there is exactly one  $z \in \mathbf{Oz}$  such that  $z \leq x < z + 1$ . This  $z = \lfloor x \rfloor_{\mathbf{Oz}}$  is parameter-free definable from  $x$  in set theory, and  $\lfloor x \rfloor_{\mathbf{Oz}} \preceq_s x$ .*

*Proof.* Write, by splitting the normal form,  $x = P + r + \varepsilon$ , where  $P \in \Pi$ ,  $r \in \mathbb{R}$ , and  $\varepsilon$  is infinitesimal. Put

$$\lfloor x \rfloor_{\mathbf{Oz}} = P + \begin{cases} r - 1, & r \in \mathbb{Z} \text{ and } \varepsilon < 0, \\ \lfloor r \rfloor, & \text{otherwise.} \end{cases} \quad (2.3)$$

A nonzero positive element of  $\Pi$  is larger than every real, since its leading exponent is positive. Hence every positive element of  $\mathbf{Oz}$  is at least 1. Formula (2.3) directly verifies the two inequalities. If two omnific integers had the required property, their nonzero difference would have absolute value less than 1, contradicting discreteness. Both the normal-form formula and the unique integer-part property are set-theoretic definitions.

For the prefix assertion (07), use the standard simplicity characterization  $z = \{z - 1 \mid z + 1\}$  for  $z \in \mathbf{Oz}$ . The inequalities place  $x$  strictly between  $z - 1$  and  $z + 1$ , and the simplest separator is a prefix of every separator.  $\square$

*Remark 2.4.* The correction when  $r$  is integral and  $\varepsilon < 0$  is essential:  $\lfloor \omega - \omega^{-1} \rfloor_{\mathbf{Oz}} = \omega - 1$ , not  $\omega$ . The proposition is classical [1, 6]; its proof is included to make subsequent definability closure entirely explicit. The floor and its correction are also treated in the omnific Diophantine report [22].

## 2.5 The imported exponential-field theorem

Source 07 uses the following classical input. The surreal exponential field is an elementary extension of the real exponential field, and it is o-minimal. If  $\epsilon$  is an epsilon number, then  $\mathbf{No}_{<\epsilon}$ , with the induced exponential, is an elementary substructure of  $\mathbf{No}$  and is closed under logarithms of positive elements. These are consequences of [3, Theorem 2.1 and Corollary 5.5]; the latter is stated even in the richer restricted-analytic exponential language. The original paper must be read with its erratum [4], as the birthday-cutoff report insists; Bournez and Guilmant prove that  $\mathbf{No}_\epsilon$  is stable under exponential and logarithm exactly when  $\epsilon$  is an epsilon number [5], which is the closure input used in theorem 10.3, and elementarity then follows from model completeness of the real exponential field with logarithm, which is what [3, Corollary 5.5] packages. Source 07 cited the original paper without the erratum; the merge adds it. Our conclusions about definable fragments use only the countable language  $\mathcal{L}_{\text{exp}}$ .

*Remark 2.5* (o-minimality inside a transitive model [merge]). Source 07 uses tacitly that  $\mathbf{No}_{\text{exp}}^M$  is o-minimal as an external structure when  $M$  is transitive. This holds formula by formula. The o-minimality of the real exponential field and the elementary equivalence of the surreal and real exponential fields are theorems of ZFC, and by uniform finiteness each formula  $\varphi(x, \bar{y})$  has a standard bound  $N_\varphi$  such that ZFC proves that every fibre  $\varphi(\mathbf{No}_{\text{exp}}, \bar{b})$  is a union of at most  $N_\varphi$  points and intervals. Applied inside  $M$ , this statement is first order with a standard numeral, so it holds externally of  $\mathbf{No}_{\text{exp}}^M$ .

**Lemma 2.6** (Classical birthday controls [07]). *If  $r\omega^a$  is a nonzero term in the normal form of  $x$ , then*

$$b(a) \leq b(x), \quad b(r\omega^a) \leq b(x).$$

*The normal-form length is at most  $b(x)$ , and for every surreal  $a$ ,*

$$b(a) \leq b(\Omega(a)) \leq \omega^{b(a)},$$

*where the power on the right is ordinal exponentiation.*

These are the estimates of [3, Lemmas 4.1–4.2 and the proof of Proposition 4.7], imported as established results rather than reproved from the sign-expansion analysis. Independently, the sign order immediately gives  $|a| \leq \underline{b(a)}$ , because the all-plus and all-minus sequences are the extremal sequences of a fixed length. For the merge, the upper bound  $b(\Omega(a)) \leq \omega^{b(a)}$  was re-derived from Gonshor’s sign expansion of  $\omega^a$ ; the exponent bound  $b(a) \leq b(x)$  was not re-checked, and [theorem 10.5](#) uses only its consequence  $a \geq -\underline{b(x)}$ .

### 3 Which notion of definability?

#### 3.1 Ambient definability with parameters

**Definition 3.1** ([04, 07]). Let  $M \models \text{ZFC}$  be a set model, and let  $A$  be an external collection of allowed elements of  $M$ . An element  $a \in M$  is *ambiently  $A$ -definable in  $M$*  if some standard first-order formula  $\varphi(v, \bar{u})$  and some finite tuple  $\bar{p}$  from  $A$  satisfy

$$M \models \forall v [\varphi(v, \bar{p}) \leftrightarrow v = a].$$

Write  $D_A^M = \text{dcl}_{(M, \in)}(A)$ , and define

$$K_A^M = D_A^M \cap \mathbf{No}^M, \quad I_A^M = D_A^M \cap \mathbf{Oz}^M, \quad O_A^M = D_A^M \cap \text{Ord}^M.$$

When  $A = \emptyset$ , say *parameter-free definable*. For a finite tuple  $p$  write  $K_p^M$ . Superscripts are omitted when the universe is fixed.

For a set model,  $D_A^M$  is an external set, definable using the external satisfaction relation of  $M$ . In particular,

$$|D_A^M| \leq \max(\aleph_0, |A|). \tag{3.1}$$

This estimate counts standard formulas and finite parameter tuples. It is external: it does not put into  $M$  a list of all the elements definable in  $M$ . Even though  $\mathbf{No}^M$  and  $K_A^M$  are sets externally,  $\mathbf{No}^M$  is a proper class internally, and  $K_A^M$  need not be represented by a set of  $M$ .

For the intended universe  $V$ , the same terminology is a metatheoretic scheme: for each particular formula we can ask whether it uniquely defines a given object. We do not add a first-order truth predicate for  $V$ , and do not assume that the full collection  $D_\emptyset^V$  is an internally available set or class.

**Lemma 3.2** (Unique-output closure [04, 07]). *Suppose a set-theoretic formula defines a unique output  $F(\bar{x})$  from each input in its domain. If every component of  $\bar{a}$  lies in  $D_A$ , then  $F(\bar{a}) \in D_A$ .*

*Proof.* Choose definitions for the finitely many components of  $\bar{a}$ . Existentially quantify those components, assert their defining formulas, and assert the graph formula for  $F$ . This is a single formula using only finitely many parameters from  $A$ , with a unique solution.  $\square$

This elementary lemma is the mechanism behind all positive closure results in the report. It does not assert that  $D_A^M \prec M$ : general set-theoretic existential statements need not have a definable witness (07).

### 3.2 Ordinal parameters

**Definition 3.3** ([04, 07]). A surreal or omnific integer is *ordinal definable* if its canonical set code is in OD. Define

$$K_{\text{HOD}} = \mathbf{No} \cap \text{OD}, \quad I_{\text{HOD}} = \mathbf{Oz} \cap \text{OD}.$$

The standard internal definition of OD says that an object is uniquely definable over some set structure  $(V_\theta, \in)$  using finitely many ordinal parameters below  $\theta$ . Reflection identifies this with the usual notion of definability in  $V$  from ordinal parameters. The extra ordinal  $\theta$  is allowed; no truth predicate for  $V$  is required. Inside a model  $M$  the same definition gives  $K_{\text{HOD}}^M = \mathbf{No}^M \cap \text{OD}^M$ .

Both OD and HOD are definable classes of ZFC, unlike a uniform truth-based relation between arbitrary formulas and their unique solutions in  $V$ . HOD is the transitive class of sets whose transitive closures, including the sets themselves, consist of ordinal-definable sets; it is an inner model of ZFC with a canonical definable well-order [10, 15]. Every ordinal is ordinal definable, using itself as an allowed parameter. This alone shows why  $K_{\text{HOD}}$  need not be a countable collection.

### 3.3 Intrinsic ordered-field definability

For a specified language  $\mathcal{L}$  on  $\mathbf{No}^M$ , *intrinsic definability* means  $\text{dcl}_{(\mathbf{No}^M, \mathcal{L})}(P)$  for a set  $P$  of surreal parameters. The quantifiers then range over surreals, and only the symbols of  $\mathcal{L}$  may be used. The three notions — ambient, ordinal and intrinsic — should never be conflated. The set-theoretic definition of  $\omega$  is parameter-free, but  $\omega$  is not definable without parameters in the pure ordered field, or even in its exponential reduct (proposition 15.1). Conversely, allowing every surreal as a parameter trivializes definability of individual surreals.

**Theorem 3.4** (Classical language contrast [04, 07]). *In the pure ordered-field language, the parameter-free definable surreals are exactly the real algebraic numbers. Their intersection with  $\mathbf{Oz}$  is exactly  $\mathbb{Z}$ . More generally, in a real closed surreal field  $F$  (for example  $F = \mathbf{No}^M$ ) and for any external parameter set  $P \subseteq F$ ,*

$$\text{dcl}_F(P) = \text{rcl}_F(\mathbb{Q}(P)),$$

*the relative real algebraic closure of  $\mathbb{Q}(P)$  in  $F$ .*

*Proof.* Quantifier elimination for real closed fields reduces a definable singleton to a Boolean combination of polynomial sign conditions over the field generated by finitely many parameters [7, 8]. If a point is transcendental over that field, it is not a zero of any of the finitely many nonzero polynomials appearing in the formula, so their signs are constant on a small interval

around it and the formula cannot isolate that point. Thus the point is algebraic over  $\mathbb{Q}(P)$ . Conversely, an algebraic element is specified as the  $k$ th real root of its polynomial, for a fixed finite  $k$ , with coefficients given by field terms in finitely many parameters. Real algebraic elements of  $\mathbf{No}$  lie in its standard real subfield. A real is omnific exactly when it is an integer, by (2.2).  $\square$

For the full proper-class field, the argument is a formula-by-formula application of the same quantifier-elimination theorem. In particular,  $\omega$  is *ambiently* parameter-free definable, but is not parameter-free definable in the pure field. Thus “the parameter-free definable omnific integers” can mean just  $\mathbb{Z}$  when definability is taken in the ordered field, but can include  $\omega$ ,  $\omega^{1/2}$  and arbitrarily elaborate bounded codes when taken in set theory; both assertions are correct in their respective languages. Nor does adding the word “omnific” to our prose add a predicate to that language. The structure  $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$  is a different structure; what the collection now proves about its definable closure is in [proposition 15.3](#).

### 3.4 Provable descriptions and computability

For a recursively axiomatized theory  $T$ , one can restrict to formulas  $\varphi$  for which  $T \vdash \exists!x (\mathbf{No}(x) \wedge \varphi(x))$ . These are *T-provably unique descriptions*, or provable names. Their interpretation in a model of  $T$  need not be invariant between different models. For example the formula which selects 0 if CH holds and 1 otherwise has a provably unique output in ZFC, but its value depends on the model (07). Transitivity, syntactic shortness, and provable uniqueness are separate issues from absoluteness; the statement “ $T$  proves uniqueness” is not a replacement for choosing the semantic universe.

A genuine effective presentation of a surreal sign sequence gives an ambient definition when the presentation uniquely determines a well-founded sequence. The converse fails already for real numbers: the characteristic real of a noncomputable but arithmetically definable set is definable. We therefore do not identify definability with any one of the computable-surreal representations catalogued by the collection, or with a numerical approximation algorithm, and we provide no algorithm deciding whether an arbitrary formula uniquely defines a surreal. When a construction is called uniform, its defining set-theoretic formula is fixed; no internal truth predicate is thereby supplied.

## 4 The field and integer part of ambiently definable numbers

Unless stated otherwise the statements of this section hold for  $V$  as a scheme and for any set model  $M$ ; the superscript  $M$  refers to such a model, whose internal operations on surreals and normal forms are used. Put  $\Pi_A = K_A \cap \Pi$  and  $R_A = K_A \cap \mathbb{R}$ .

**Theorem 4.1** (Definable field and integer part [04, 07]). *The ambiently  $A$ -definable surreals  $K_A$  form a real closed field. It is closed under  $\Omega$ , under  $\Omega^{-1}$  on monomials, under  $\exp$  and positive logarithms, and under the omnific floor. The ring  $I_A = K_A \cap \mathbf{Oz}$  is a discretely ordered integer part of  $K_A$ , and*

$$K_A = \text{Frac}(I_A). \quad (4.1)$$

*More precisely, every nonzero  $x \in K_A$  can be written  $x = p/q$  with  $p, q \in \Pi_A$  and  $q > 0$  a monomial, the pair  $(p, q)$  being chosen by one parameter-free definable rule from  $x$ ; two such rules are*

$$q = \omega^\delta \quad (04), \quad q = \omega^b, \quad b = \{ \{0\} \cup \{-a : a \in \text{supp } x\} \mid \emptyset \} \quad (07), \quad (4.2)$$

*where  $\delta$  is the least positive ordinal with  $\delta + g > 0$  for all  $g \in \text{supp } x$ . The field  $K_A$  is an elementary substructure of the ambient surreal ordered field in the pure ordered-field language.*

*Proof.* The constants  $0, 1$ , addition, subtraction, multiplication, inverse away from zero,  $\Omega$ ,  $\Omega^{-1}$ ,  $\exp$  and  $\log$  are uniquely definable set-theoretic operations, so [lemma 3.2](#) proves the field axioms and the closure assertions by restriction. Every positive element has a unique positive square root in  $\mathbf{No}$ , so that square root belongs to  $K_A$ . For an odd-degree polynomial with coefficients in  $K_A$ , real closedness of  $\mathbf{No}$  gives at least one root; its finite set of roots has a least member, uniquely definable from the coefficient tuple. That root lies in  $K_A$ . The square-root and odd-degree-root criterion proves real closedness. This is an application of the classical real-closed-field theorem, not a new proof of real closedness of  $\mathbf{No}$ .

The floor from [proposition 2.3](#) belongs to  $D_A$  whenever its input does. Hence every  $x \in K_A$  has its omnific floor in  $I_A$ . Discreteness and the subring property restrict from  $\mathbf{Oz}$ .

For the fraction-field assertion, let  $0 \neq x \in K_A$ , and put  $S = \text{supp}(x)$ . The whole normal form and  $S$  are definable from  $x$ . By [lemma 2.1](#) there exists a positive ordinal  $\delta$  with  $\delta + g > 0$  for  $g \in S$ . Choose the *least* such positive ordinal (04). It is definable from  $x$ , so  $\delta \in K_A$ , and  $q = \omega^\delta$  and  $p = qx$  belong to  $K_A$ . Alternatively (07), form the set cut  $b$  in (4.2); then  $b > 0$  and  $b + a > 0$  for every  $a \in S$ , and  $b$  is definable from  $x$ . In either case the normal form of  $p = qx$  is obtained by adding the exponent of  $q$  to every exponent of  $x$ , without changing the coefficients. Its exponents are strictly positive, so  $p, q \in \Pi_A$ , and  $x = p/q$ . The zero case is  $0 = 0/1$ . The reverse inclusion follows because  $K_A$  is a field. Finally, every inclusion of real closed fields is elementary by quantifier elimination [7].  $\square$

*Remark 4.2.* The proof does not suppose that each exponent of an  $A$ -definable normal form is  $A$ -definable. It only uses definability of the *entire support* to choose one bound. Confusing these two assertions would invalidate the fraction-field argument. The equality  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and the support-clearing method are not new: they are `odg:thm:fractions` in the omnific Diophantine report [22]. The point here is that the denominator can be chosen canonically enough to preserve the *same* allowed parameters; merely invoking an unspecified denominator would not establish this conclusion.

Source 07 states the closure under  $\Omega$ ,  $\exp$  and the floor, and the cut denominator, for a transitive model; the proofs use only [lemma 3.2](#), so they hold in the generality of 04's statement. The same applies to [proposition 4.9](#) and [theorem 4.10](#) below.

**Lemma 4.3** (Definable selection in an o-minimal field [07]). *Let  $F$  be an o-minimal expansion of an ordered field. For each formula  $\varphi(y, \bar{x})$  there is a parameter-free definable partial function  $s_\varphi(\bar{x})$  whose domain is exactly*

$$\{\bar{a} : F \models \exists y \varphi(y, \bar{a})\}$$

*and whose value satisfies  $\varphi$ . A substructure closed under these functions is elementary in  $F$ . Intersections of substructures closed under all of them are elementary as well.*

*Proof.* A nonempty definable subset of the line is a finite union of points and intervals. Its leftmost component and its finite endpoints are definable from its parameters. Select its unique point if it is a point. For an interval choose its left endpoint if that endpoint belongs to it; otherwise use the midpoint when both endpoints are finite, the finite endpoint minus or plus 1 in the one-sided unbounded cases, and 0 for the whole line. The alternatives and endpoints are uniformly definable for the given formula. Thus this is a definable selection, not a choice made separately with new parameters for each fibre.

Apply the Tarski-Vaught criterion to the selected witness. One-coordinate selection suffices for all existential formulas; tuples can be handled successively. Closure of an intersection follows because the same single-valued selectors are used in every substructure.  $\square$

**Theorem 4.4** (Exponential elementarity [07]). *Let  $M$  be a transitive set model of ZFC. In the ordered exponential language,*

$$(K_A^M, +, \cdot, <, \exp) \prec (\mathbf{No}^M, +, \cdot, <, \exp).$$

*In particular the same holds in the ordered-field reduct.*

*Proof.* By section 2.5 and remark 2.5 the right-hand structure is o-minimal. Each fixed  $\mathcal{L}_{\exp}$ -formula translates into a fixed formula of set theory: restrict its quantifiers to  $\mathbf{No}$ , and replace its function symbols by their definable graphs. Thus a selector of lemma 4.3, evaluated at a tuple from  $K_A^M$ , is an ambient uniquely definable output. It lies in  $K_A^M$  by lemma 3.2. The selection lemma gives elementarity. This translation is performed for each standard formula separately. The proof never combines all translations into a truth predicate inside  $M$ .  $\square$

*Remark 4.5.* A language with a symbol for every restricted real analytic function can contain nondefinable real information in its symbols. The countability and ambient-definability statements here concern  $\mathcal{L}_{\exp}$ , not that arbitrary uncountable expansion (07).

**Corollary 4.6** (Open induction [04]). *The nonnegative part of  $I_A$  satisfies induction for every quantifier-free formula in the ordered-ring language, with parameters from  $I_A$ .*

*Proof.* A quantifier-free formula with parameters in  $I_A$  defines a semialgebraic subset of the real closed field  $K_A$ , hence a finite union of intervals and points. Every nonempty intersection of such a set with  $(I_A)_{\geq 0}$  has a least member. Indeed, the integer-part property supplies the first integer at or immediately above each finite left endpoint, with a one-unit adjustment for a strict endpoint; an interval unbounded to the left is treated by the lower bound 0. Take the least of the finitely many nonempty pieces. If an induction instance failed, the set of counterexamples would therefore have a least member. It is not 0, and its predecessor is nonnegative and satisfies the formula, contradicting the inductive step. This is the standard integer-part route to open induction [11, 6].  $\square$

For the full ring  $\mathbf{Oz}$  open induction, its failure beyond open formulas, and the failure of Peano arithmetic are in the omnific Diophantine report (`odg:rem:foursquares`, `odg:thm:induction`); the corollary is the definable-fragment instance.

Related (batch 86, added 3 October 2026): open induction is sharp here. By proposition 16.1,  $I_A$  is a discretely ordered ring with division with remainder by every ordinary integer and with the ring retraction `ct` onto  $\mathbb{Z}$ . The discrete-initial-subgroups report (its source 03; AI-assisted, unrefereed, not formalized) shows that such a ring satisfies the single bounded induction instance of its dyadic standard-cut formula  $\text{Std}^{\text{dy}}$  only if it is  $\mathbb{Z}$  (`isg:sc:thm:boundary`); so the nonnegative part of  $I_A$  satisfies open induction but not  $\text{!}\Delta_0$ , as for  $\mathbf{Oz}$ . (That report proves its theorems for set-sized rings and reads class applications as in `isg:sc:sec:classes`; the same reading applies to  $I_A$ .) The `polish-models-of-omnific-arithmetic` report proves open induction by the route of the corollary above for two finite principal-part integer parts:  $\mathbb{Z} \oplus \bigoplus_{q>0} \mathbb{R}X^q$ , in a Borel presentation (`pma:bor:thm:arithmetic`), and its countable subring with real algebraic coefficients, inside a Polish ordered field (`pma:ser:prop:iopen`).

**Proposition 4.7** (Finite ambiguity does not enlarge the hull [04]). *An element of  $\mathbf{No}$  belonging to a finite ambiently  $A$ -definable set of surreals is itself ambiently  $A$ -definable. In particular, ambient model-theoretic algebraicity and definability coincide on the surreal sort. Every surreal algebraic over  $K_A$  already belongs to  $K_A$ .*

*Proof.* Order the finite candidate set by the surreal order. Each member is its  $k$ th member for a fixed finite  $k$ ; this property is expressible by a formula with the same parameters. For

the final assertion, the roots of a nonzero polynomial over  $K_A$  form such a finite definable set. Equivalently, a real closed field has no proper algebraic ordered extension.  $\square$

This finite-ordering argument is a particularly transparent case of model-theoretic algebraicity; broader set-theoretic questions about algebraicity and implicit definability are studied in Hamkins–Leahy [14]. It concerns a genuinely finite candidate set, not an internally nonstandard finite set viewed from outside a nonstandard model.

**Proposition 4.8** (Whole-family closure [04, 07]). *If  $L, R \in D_A$  are sets of surreals with  $L < R$ , their simplest separator belongs to  $K_A$ . If an entire set-coded, strongly summable family of surreals belongs to  $D_A$  — in particular a valid normal form whose whole indexed sequence is  $A$ -definable — its canonical sum belongs to  $K_A$ .*

*Proof.* In both cases the output is a unique parameter-free definable function of the input set codes; apply lemma 3.2. For a cut, both option sets must be jointly available as  $A$ -definable sets, not merely have individually  $A$ -definable elements. Strong summability is the normal-form condition that the union of supports is reverse well ordered and each exponent receives only finitely many nonzero contributions.  $\square$

The hypothesis concerns the *whole input code*. An arbitrary set of individually definable entries need not satisfy it (section 9). This distinction is central, not a technical qualification at the edge of the theory.

**Proposition 4.9** (The normal-form invariant [07]). *For every  $x \in \mathbf{No}^M$ ,*

$$x \in K_A^M \iff \text{NF}(x) \in D_A^M.$$

*If  $x \in K_A^M$  and  $\xi < \text{dom NF}(x)$  is an  $A$ -definable ordinal, then  $g_\xi$  and  $r_\xi$  are  $A$ -definable.*

*Proof.* The normal-form map and its inverse are unique parameter-free definable constructions. Evaluation of a set-coded function at an input is also uniquely definable. Apply lemma 3.2 in both directions.  $\square$

Without the definability of the index, the final assertion need not hold (proposition 9.5 and theorem 9.6). Without the definability of the *whole* graph, its converse need not hold either (theorems 9.1 and 9.3).

## 4.1 Value group and residue field

Define  $v(x) = -g_0$  for nonzero  $x$  in (2.1), and  $v(0) = \infty$ . This is the normal-form valuation; it uses the negative of the largest Conway exponent. Let

$$\mathcal{O}_A = \{x \in K_A : |x| < n \text{ for some } n \in \mathbb{N}_{>0}\}, \quad \mathfrak{m}_A = \{x \in K_A : |x| < 1/n \text{ for every } n \in \mathbb{N}_{>0}\}.$$

Here the quantification over  $\mathbb{N}$  is ambient set-theoretic, not a claim that  $\mathbb{N}$  is intrinsically definable in an o-minimal field.

**Theorem 4.10** (Value group and residue field [07]). *The value group of  $K_A$  is canonically the ordered additive group  $(K_A, +)$ :  $v(K_A^\times) = K_A$ . Standard part induces an isomorphism  $\mathcal{O}_A/\mathfrak{m}_A \simeq R_A$ . Every finite  $x \in K_A$  has a unique decomposition  $x = r + \varepsilon$  with  $r \in R_A$  and  $\varepsilon \in \mathfrak{m}_A$ .*

*Proof.* The leading exponent is a unique definable function of a nonzero normal form, so  $v(x) \in K_A$ . Conversely  $v(\omega^{-a}) = a$  for every  $a \in K_A$ , and the monomial lies in  $K_A$ .

For finite  $x$ , its real constant term is its standard part; this remains true when there is no constant term, in which case it is zero. This unique real is definable from  $x$ , hence belongs to  $R_A$ . The ordinary normal-form multiplication and addition laws show that standard part is a surjective ring homomorphism with kernel exactly the infinitesimals. Uniqueness of the decomposition follows because no nonzero real is infinitesimal.  $\square$

This self-similarity of the value group should not be mistaken for a claim that  $K_A$  contains every Hahn series whose entries belong to  $K_A$ ; [section 9](#) gives explicit obstructions.

## 5 The ordinal-definable surreal field is the HOD field

The canonical sign representation makes hereditary definability much more transparent than it is for arbitrary sets. In this section  $V$  is the universe of construction; every statement can equally be read inside a model, with  $V$ , OD and HOD relativized.

**Theorem 5.1** (HOD identification [04, 07]). *With canonical sign-sequence representatives,*

$$K_{\text{HOD}} = \mathbf{No} \cap \text{OD} = \mathbf{No} \cap \text{HOD} = \mathbf{No}^{\text{HOD}}, \quad I_{\text{HOD}} = \mathbf{Oz} \cap \text{OD} = \mathbf{Oz} \cap \text{HOD} = \mathbf{Oz}^{\text{HOD}}. \quad (5.1)$$

*These are equalities of sign-sequence classes. The class  $K_{\text{HOD}}$  is an initial subclass of  $\mathbf{No}$  and a real closed field that contains every ordinal and is closed under  $\Omega$ ,  $\Omega^{-1}$  on monomials,  $\exp$ , positive logarithms and the omnific floor. The ring  $I_{\text{HOD}}$  is an integer part with  $\text{Frac}(I_{\text{HOD}}) = K_{\text{HOD}}$ .*

*Proof.* Let  $s : \alpha \rightarrow \{-, +\}$  be ordinal definable. Every element in its transitive closure, other than  $s$  itself, is built from ordinals and the two fixed sign symbols by finitely many fixed set constructions, so it is ordinal definable. Since  $s$  itself is ordinal definable,  $s \in \text{HOD}$ . The converse is part of the definition of HOD. HOD contains every ordinal and is transitive, so being an ordinal-length sign sequence is absolute, and its surreal sign sequences are exactly the ambient sign sequences that belong to HOD. This gives the carrier equalities for  $\mathbf{No}$  in (5.1).

The closure assertions follow by substituting ordinal definitions into the unique-output constructions of [theorem 4.1](#); the ordinals used are permitted parameters. The floor and the least-ordinal denominator give the integer-part and fraction claims for  $I_{\text{HOD}} = \mathbf{Oz} \cap \text{OD}$ . Initiality: a prefix  $x \upharpoonright \beta$  is definable from  $x$  and the ordinal  $\beta$  (07).

It remains to compare with the internal arithmetic of HOD. Comparison is computed from the same signs; canonical options are the same initial segments; and the recursive arithmetic on set-sized option closures is absolute between a transitive inner model and the ambient universe, so the ordered-field structure of  $\mathbf{No}^{\text{HOD}}$  is the restriction of that of  $\mathbf{No}$  (this part is `univ:prop:absolute` in the universes report, for any same-ordinal inner model). For the monomial map one may use its recursive cut with integer and dyadic multipliers, so no new reals enter the recursion (04); source 07 asserts the same compatibility for the normal form, the exponential and the purely infinite/constant splitting. The universes report proves the arithmetic and cut parts only, so for these further operations the recursion argument just sketched is the source. For the omnific carriers, an OD surreal has an OD normal-form graph whose entries at ordinal indices are OD, hence the whole graph is in HOD ([lemma 5.2](#)) and its canonical interpretation agrees in the two universes; this gives  $\mathbf{Oz} \cap \text{HOD} = \mathbf{Oz}^{\text{HOD}}$  (07).  $\square$

When  $V$  is replaced by a transitive model  $M$ , [theorem 4.4](#) with all ordinals of  $M$  as the external parameter set also shows that  $K_{\text{HOD}}^M$  is an elementary exponential subfield of  $\mathbf{No}^M$  (07). This is stronger than merely being closed under  $\exp$ .

**Lemma 5.2** (Support of an ordinal-definable number [04, 07]). *If  $x \in K_{\text{HOD}}$  has normal form (2.1), then for every  $\xi < \lambda$ ,*

$$r_\xi \in \mathbb{R} \cap \text{HOD}, \quad g_\xi \in K_{\text{HOD}}.$$

*In fact,  $\text{NF}(x) \in \text{HOD}$ , and*

$$x \in K_{\text{HOD}} \iff \text{NF}(x) \in \text{HOD}. \quad (5.2)$$

*The converse with only “every coefficient and exponent is OD” is false whenever there is a non-OD subset of an ordinal.*

*Proof.* The entire normal form is uniquely definable from  $x$ , so is ordinal definable. An individual entry is uniquely definable from  $x$  and the ordinal index  $\xi$ . Its real coefficient is ordinal definable, hence hereditarily ordinal definable under the usual real coding. Its surreal exponent belongs to HOD by theorem 5.1. The length, indices, and the fixed tuple coding are also hereditarily ordinal definable. Hence the whole sequence belongs to HOD. Conversely,  $\text{NF}(x) \in \text{HOD}$  implies that it is ordinal definable, so its unique surreal value is ordinal definable. The counterexamples are theorem 9.1 and  $E_\alpha(X)$  of theorem 9.3 for a non-OD  $X \subseteq \alpha$ .  $\square$

The last equivalence requires the whole sequence to lie in HOD. The strictly weaker condition “every coefficient and exponent lies in HOD” is insufficient, even with a HOD support and rational coefficients. In particular,  $K_{\text{HOD}}$  is not being identified with the unrestricted ambient Hahn field containing *all* externally assembled series with coefficients and exponents in HOD.

**Corollary 5.3** (No algebraic defect [04]). *Every  $x \in \mathbf{No} \setminus K_{\text{HOD}}$  is transcendental over  $K_{\text{HOD}}$ . The inclusion  $K_{\text{HOD}} \subseteq \mathbf{No}$  is elementary in the ordered-field language, whether or not  $V = \text{HOD}$ .*

*Proof.* Apply proposition 4.7 with ordinal parameters, and then quantifier elimination.  $\square$

**Proposition 5.4** (The HOD field is not intrinsically definable when proper [04]; proof corrected [merge]). *If  $K_{\text{HOD}} \neq \mathbf{No}$ , no ordered-field formula, even with finitely many surreal parameters, defines  $K_{\text{HOD}}$  inside  $\mathbf{No}$ .*

*Proof.* It suffices to show that an infinite definable subfield  $F$  is the whole field;  $K_{\text{HOD}}$  is infinite. An infinite definable subset of a real closed field is a finite union of points and intervals, so  $F$  contains an open interval  $(a, b)$ . The endpoints need not lie in  $F$ , so choose  $a' < b'$  inside  $(a, b)$ ; they belong to  $F$ , and so does  $c = b' - a' > 0$ . For  $0 < y < c$  the point  $a' + y$  lies in  $(a', b') \subseteq F$ , so  $y \in F$ ; hence  $(0, c) \subseteq F$ . For any  $x > 0$ , the number  $y = 1/(x + 2/c)$  lies in  $(0, c)$ , hence in  $F$ , and then  $x = 1/y - 2/c \in F$ . Negatives and zero follow, so  $F$  is the entire field.  $\square$

Source 04 took  $c$  from “differences” of the interval without placing the endpoints in the subfield; the choice of  $a', b'$  inside the interval closes that gap. The same fact, that a proper real closed subfield has empty interior, is used in proposition 7.6.

## 6 Reversible omnific codes

### 6.1 A sign code in a fixed omnific interval [04]

For each ordinal  $\beta$ , viewed as a surreal ordinal, define

$$e_\beta = \omega^{-(\beta+1)}. \quad (6.1)$$

The  $e_\beta$  form a strictly decreasing class of positive infinitesimal surreals. In particular  $0 < e_\beta \leq \omega^{-1} < 1/2$ . Every  $e_\beta$  is ordinal definable. For a sign sequence  $s : \alpha \rightarrow \{-, +\}$  put

$$c_s(\beta) = \begin{cases} 1, & s(\beta) = -, \\ 2, & s(\beta) = +. \end{cases}$$

**Definition 6.1** (Self-delimiting omnific code [04]). Define

$$\mathbf{C}(s) = \omega + \sum_{\beta < \alpha} c_s(\beta) \omega^{e_\beta} + 3\omega^{e_\alpha}, \quad \alpha = \mathbf{b}(s). \quad (6.2)$$

The coefficient 3 is an end marker, not a third sign value.

**Theorem 6.2** (Reversible fixed-leading-term coding [04]). *Formula (6.2) defines a parameter-free set-theoretic injection  $\mathbf{C} : \mathbf{No} \rightarrow \mathcal{U}$ . Its image is set-theoretically definable, and there is a parameter-free definable inverse  $\mathbf{D}$  on that image. For every model  $M$  and every allowed parameter family  $A$ ,*

$$s \in D_A^M \iff \mathbf{C}^M(s) \in D_A^M. \quad (6.3)$$

*In particular,  $s$  and  $\mathbf{C}(s)$  are mutually definable and have the same ordinal-definability status.*

*Proof.* The exponents in (6.2) occur in strictly decreasing order: first 1, then  $e_\beta$  for  $\beta \leq \alpha$ . They form a set, because  $\alpha$  is an ordinal. Each coefficient is nonzero and real, so the expression is a valid Conway normal form. Every exponent is positive. Thus the value is an omnific integer with constant term zero. Its leading term is exactly  $\omega$ .

The tail  $\mathbf{C}(s) - \omega$  is positive. Its leading exponent is  $e_0 = \omega^{-1}$ , including the empty-sign case, in which the leading tail coefficient is 3. Since  $e_0 < 1/2$ , comparison of normal forms gives  $\mathbf{C}(s) - \omega < \omega^{1/2}$ . Hence  $\mathbf{C}(s) \in \mathcal{U}$ .

To decode, take the unique normal form of  $z$ . Verify that its leading term is  $\omega$ , that its remaining exponents are exactly  $e_\beta$  for an initial ordinal segment  $\beta \leq \alpha$ , that the last coefficient is 3, and that every preceding coefficient is 1 or 2. The map  $\beta \mapsto e_\beta$  is injective, so the end marker uniquely determines  $\alpha$ . At each  $\beta < \alpha$ , recover  $-$  from coefficient 1 and  $+$  from coefficient 2. This produces exactly one sign sequence  $s$  and satisfies  $\mathbf{D}(\mathbf{C}(s)) = s$ .

These are set-theoretic definitions: the assertion that there exists an ordinal  $\alpha$  and the specified indexed normal form is a first-order assertion about sets. It neither quantifies over class sequences nor needs a field-intrinsic way to recognize  $\omega$ . The unique-output closure lemma applied to both maps proves (6.3). The same proof is a theorem internal to any model; external definability uses standard formulas for the two maps.  $\square$

**Example 6.3.** The first three codes are

$$\mathbf{C}(0) = \omega + 3\omega^{\omega^{-1}}, \quad \mathbf{C}(1) = \omega + 2\omega^{\omega^{-1}} + 3\omega^{\omega^{-2}}, \quad \mathbf{C}(-1) = \omega + \omega^{\omega^{-1}} + 3\omega^{\omega^{-2}}.$$

The code for 0 is not 0. The coding theorem makes no assertion of arithmetic compatibility.

**Corollary 6.4** (Congruence and leading-data blindness [04]). *Every code has leading exponent 1, leading coefficient 1, constant term 0, and coefficients from  $\{1, 2, 3\}$ . Moreover, for every ordinary positive integer  $n$ ,*

$$\mathbf{C}(s) \in n\mathbf{Oz}.$$

*These invariants therefore do not bound ambient definability complexity.*

*Proof.* Division by an ordinary  $n > 0$  scales the real coefficients and leaves all exponents positive, so the quotient is still omnific. The remaining assertions follow from the normal form. Since every surreal is recoverable from its code, the listed common invariants cannot separate definability degrees.  $\square$

**Proposition 6.5** (A complementary two-term code [04]). *There is also a parameter-free definable order isomorphism*

$$\mathsf{T} : \mathbf{No} \longrightarrow \mathcal{U}_2 := \{\omega + \omega^g : 0 < g < 1/2\} \subseteq \mathcal{U}.$$

*Every value has exactly two nonzero normal-form terms, both with coefficient 1, and is mutually ambiently definable with its input.*

*Proof.* The rationally defined map

$$h(x) = \frac{1}{4} \left( 1 + \frac{x}{1 + |x|} \right)$$

is an increasing bijection from  $\mathbf{No}$  onto  $(0, 1/2)$ . For its inverse, put  $u = 4g - 1$  and use  $x = u/(1 - |u|)$ . Therefore  $\mathsf{T}(x) = \omega + \omega^{h(x)}$  has the stated normal form, is strictly increasing, and has exactly the stated image. Extracting the lower exponent from the unique normal form and applying the inverse of  $h$  recovers  $x$  by a parameter-free set-theoretic operation.  $\square$

The two codes have different purposes. The elementary two-term code allows an arbitrary surreal exponent to carry all the input information. The sign code  $\mathsf{C}$  instead fixes its exponents in an ordinal-definable template, so that all information is stored in the assignment of a finite coefficient alphabet. It is the latter feature that proves the failure of termwise HOD closure (theorem 9.1).

*Remark 6.6* (An interval is still a proper class [04]). Although  $\mathcal{U}$  is bounded by two explicitly given surreals, it is a proper class in the ambient universe. Otherwise its subset  $\text{ran}(\mathsf{C})$  would be a set and Replacement applied to  $\mathsf{D}$  would make all of  $\mathbf{No}$  a set. Ordered boundedness is not a size bound for the full surreal field.

## 6.2 A monomial code below omega [07]

**Lemma 6.7** ([07]). *In every real closed ordered field  $F$ , the map*

$$\tau(x) = \frac{1}{2} \left( 1 + \frac{x}{\sqrt{1 + x^2}} \right)$$

*is an order isomorphism  $F \rightarrow (0, 1)_F$ . Its inverse is*

$$\tau^{-1}(t) = \frac{2t - 1}{\sqrt{1 - (2t - 1)^2}}. \tag{6.4}$$

*Both maps are parameter-free definable in the ordered-field language.*

*Proof.* Since  $1 + x^2 > x^2$ , we have  $|x| < \sqrt{1 + x^2}$ , so  $0 < \tau(x) < 1$ . If  $u \in (-1, 1)$ , put  $x = u/\sqrt{1 - u^2}$ . Direct calculation gives

$$1 + x^2 = \frac{1}{1 - u^2}, \quad \frac{x}{\sqrt{1 + x^2}} = u.$$

This proves the inverse formula. To check monotonicity without analytic assumptions, the function  $x/\sqrt{1 + x^2}$  preserves sign. For  $0 \leq x < y$ , comparison of its squares reduces to

$$x^2(1 + y^2) < y^2(1 + x^2),$$

which is equivalent to  $x^2 < y^2$ . The negative case follows by oddness, and opposite signs are immediate.  $\square$

**Theorem 6.8** (Bounded omnific coding [07]). *The function*

$$Q(x) = \Omega(\tau(x)) = \omega^{\tau(x)} \quad (6.5)$$

*is a parameter-free ambient-definable order embedding of  $\mathbf{No}$  into  $\mathbf{Oz} \cap (0, \omega)$ . Its image is exactly  $\{\omega^a : 0 < a < 1\}$ , and its inverse there is  $\tau^{-1} \circ \Omega^{-1}$ . For every positive ordinary integer  $n$ ,*

$$n < Q(x) < \omega/n. \quad (6.6)$$

*For any  $M$  and  $A$  as in [definition 3.1](#),*

$$x \in K_A^M \iff Q(x) \in I_A^M. \quad (6.7)$$

*More strongly,  $\text{dcl}_{(M, \in)}(x) = \text{dcl}_{(M, \in)}(Q(x))$ .*

*Proof.* The exponent  $\tau(x)$  lies strictly between 0 and 1. A one-term normal form with a positive exponent is purely infinite, hence omnific. Comparison of the exponents 0,  $\tau(x)$ , and 1 gives (6.6). Both  $\tau$  and  $\Omega$  are increasing, so their composite is an order embedding. The displayed inverse follows from uniqueness of the monomial exponent and [lemma 6.7](#). Each direction is a unique ambient definition, so [lemma 3.2](#) gives (6.7). Substitution of these definitions also gives the equality of definable closures, with or without additional parameters.  $\square$

This theorem does not assert that the interval is order-isomorphic to all of  $\mathbf{No}$ : the image consists only of selected monomials. Nor does it give an ordered-field-definable map; such a map into this discrete set would contradict o-minimality. Its use of the ambient omega map is indispensable. The code  $Q$  and 04's two-term code  $T$  rest on the same idea, an algebraic compression of  $\mathbf{No}$  into an interval followed by the omega map;  $Q$  has a single monomial term and lands below  $\omega$ ,  $T$  has two terms and lands in  $\mathcal{U}$ .

**Theorem 6.9** (Local omnific coding [07]). *Let  $a < b$  be omnific integers. If  $\ell = b - a$  is positive infinite and  $\beta > 0$  is its leading normal-form exponent, then*

$$Q_{a,b}(x) = a + \omega^{\beta\tau(x)} \quad (6.8)$$

*is an order embedding  $\mathbf{No} \rightarrow (a, b) \cap \mathbf{Oz}$ , ambient-definable over  $a, b$ , with an inverse definable over the same parameters. Consequently  $x$  and  $Q_{a,b}(x)$  are interdefinable over  $a, b$ . If  $b - a$  is finite,  $(a, b) \cap \mathbf{Oz}$  is finite, so no injection from all surreals into it exists.*

*Proof.* The leading coefficient of  $\ell$  is positive. Since  $0 < \beta\tau(x) < \beta$ , the monomial in (6.8) is positive infinite but of strictly smaller Archimedean order than  $\ell$ . It is therefore strictly between 0 and  $\ell$ . The sum is omnific, and monotonicity follows from  $\beta > 0$ . For a point  $z$  in the image, recover  $\gamma = \Omega^{-1}(z - a)$ , then  $\tau(x) = \gamma/\beta$ , then  $x$  from (6.4). The leading exponent  $\beta$  is definable from  $a, b$ , so no additional parameter was introduced. If  $\ell$  is finite it is an ordinary positive integer by (2.2). Any omnific difference strictly between 0 and  $\ell$  is then one of  $1, \dots, \ell - 1$ , so the interval contains exactly those finitely many translates of  $a$ .  $\square$

The relevant condition is infinite *width*, not infinite endpoints. For example  $(\omega, \omega + 3)$  contains just two omnific integers, while  $(\omega, 2\omega)$  can encode every surreal by the theorem. Compare [corollary 8.5](#), which places an independent family at every definable infinite scale.

### 6.3 A uniformly definable bounded family with unbounded birthdays [07]

An order embedding into a numerically bounded interval need not preserve birthdays. Nevertheless its definable inverse lets us force birthdays to escape every prescribed ordinal bound.

**Theorem 6.10** (Canonical birthday escape [07]). *There is a parameter-free set-theoretically definable class function  $\text{esc} : \text{Ord} \rightarrow \mathbf{Oz} \cap (0, \omega)$  such that  $\text{b}(\text{esc}(\alpha)) > \alpha$  for every ordinal  $\alpha$ . It requires no global choice and no truth predicate.*

*Proof.* For a fixed ordinal  $\alpha$ , the class  $\mathbf{No}_{\leq \alpha} = \{z \in \mathbf{No} : \text{b}(z) \leq \alpha\}$  is a set: its members are sign sequences of bounded ordinal length. By Separation and Replacement, the inverse image

$$Q_\alpha = \{Q^{-1}(z) : z \in \mathbf{No}_{\leq \alpha} \cap \text{ran}(Q)\}$$

is a set. Define

$$u_\alpha = \{Q_\alpha \mid \emptyset\}, \quad \text{esc}(\alpha) = Q(u_\alpha). \quad (6.9)$$

Every construction is uniquely and uniformly definable from  $\alpha$ . Suppose  $\text{b}(\text{esc}(\alpha)) \leq \alpha$ . Then  $\text{esc}(\alpha) \in \mathbf{No}_{\leq \alpha} \cap \text{ran}(Q)$ , so its unique inverse  $u_\alpha$  belongs to  $Q_\alpha$ . But the defining cut makes  $u_\alpha$  strictly greater than every member of  $Q_\alpha$ , including itself. This contradiction proves the strict birthday bound.  $\square$

**Corollary 6.11** ([07]). *For every  $A$ -definable ordinal  $\alpha$  of  $M$ , there is an  $A$ -definable omnific integer  $z$  with  $0 < z < \omega$  and  $\text{b}(z) > \alpha$ . The ordinal-definable omnific integers in  $(0, \omega)$  have birthdays unbounded in  $\text{Ord}$  in every universe of ZFC, independently of whether  $V = \text{HOD}$ .*

*Proof.* Apply [theorem 6.10](#) internally and [lemma 3.2](#). With ordinal parameters allowed, every  $\text{esc}(\alpha)$  is ordinal-definable.  $\square$

The last statement does not imply density of ordinal-definable surreals. Birthday cofinality of a bounded family and density in the numerical order are different properties; Hamkins's density theorem has the substantially stronger hypothesis discussed in [theorem 7.5](#). The escape  $\text{esc}$  is also different from 04's  $\Sigma_n$  escape of [theorem 12.3](#), which escapes a definability level rather than a birthday.

## 7 One interval detects the entire set-theoretic universe

We first spell out the standard set-coding step needed for the global consequences. This step is not an original bi-interpretation claim.

**Lemma 7.1** (Every set is recoverable from some surreal [04, 07]). *There is a parameter-free partial set-theoretic function  $\text{Set}$  on surreal sign sequences such that every set is  $\text{Set}(s)$  for some  $s$ . No parameter-free selection of such an  $s$  from the set being coded is asserted. In particular, if every surreal of a model  $M$  is parameter-free definable over  $M$ , then every set of  $M$  is.*

*Proof.* For a set  $a$ , Choice gives a bijection  $f : \kappa \rightarrow \text{TC}(\{a\})$  for some ordinal  $\kappa$ . Transport membership to a well-founded extensional relation  $R$  on  $\kappa$ , and record the unique index  $\varrho$  with  $f(\varrho) = a$ .

Fix a parameter-free class pairing of ordinals  $p : \text{Ord}^2 \rightarrow \text{Ord}$ . One construction well-orders ordinal pairs first by their maximum and then lexicographically, and uses the order type of each predecessor set. Every predecessor collection is a set, so this defines a class bijection. (07 uses instead the injection  $(\xi, \eta) \mapsto \kappa \cdot \xi + \eta$  with ordinal arithmetic.) Form the set of ordinals

$$X_a = \{p(0, \kappa), p(1, \varrho)\} \cup \{p(2, p(\xi, \eta)) : (\xi, \eta) \in R\}.$$

Let  $\lambda = \sup\{\gamma + 1 : \gamma \in X_a\}$  and take the sign sequence of length  $\lambda$  whose positive positions are exactly  $X_a$ . From this sequence recover  $X_a$ , then  $\kappa$ ,  $\varrho$ ,  $R$ , and finally the value at  $\varrho$  of the unique Mostowski collapse. These operations are parameter-free and unique whenever the code is valid. For invalid codes leave  $\text{Set}$  undefined. The last sentence follows by [lemma 3.2](#) applied inside  $M$ . Mostowski collapse is standard [\[10\]](#); the same pointed-graph mechanism is present in the announced surreal/set-theory dictionary [\[17\]](#).  $\square$

The use of Choice in this proof is existential. It does *not* produce a canonical definable well-ordering of every transitive closure from its underlying set. In particular, the lemma must not be strengthened to a definable injection of arbitrary sets into omnific integers without further assumptions (07).

**Theorem 7.2** (The ordinal-definability test [\[04, 07\]](#)). *The following statements are equivalent in ZFC:*

- (i)  $V = \text{HOD}$ ;
- (ii) *every surreal is ordinal definable*;
- (iii) *every omnific integer is ordinal definable*;
- (iv) *every member of  $\mathcal{U}$  is ordinal definable* [\[04\]](#);
- (v) *every value of the code  $\mathbf{C}$  is ordinal definable* [\[04\]](#);
- (vi) *every omnific integer in  $(0, \omega)$  is ordinal definable* [\[07\]](#);
- (vii) *every monomial  $\omega^a$  with  $0 < a < 1$  is ordinal definable* [\[07\]](#).

*Proof.* The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv)  $\Rightarrow$ (v) and (iii) $\Rightarrow$ (vi) $\Rightarrow$ (vii) are immediate from the inclusions. If (v) holds, recover each  $s$  from  $\mathbf{C}(s)$  by [theorem 6.2](#); if (vii) holds, recover each  $x$  from  $\mathbf{Q}(x)$  by [theorem 6.8](#). Either way every surreal is ordinal definable. For any set  $a$ , choose a surreal  $s$  with  $\text{Set}(s) = a$  by [lemma 7.1](#). Its canonical sign code lies in  $\text{HOD}$  by [theorem 5.1](#). More explicitly, the associated ordinal set  $X_a$  lies in  $\text{HOD}$ , and  $\text{HOD}$  decodes the same well-founded extensional relation and its Mostowski collapse. Thus  $a \in \text{HOD}$ . Since  $a$  was arbitrary,  $V = \text{HOD}$ .  $\square$

The equivalence of (i) and (ii) is in substance part of Hamkins’s 2012 answer ([theorem 7.5](#)); source 04 did not cite it. The bounded conditions (iv)–(vii) are the forms proposed by the two sources.

**Corollary 7.3** (Two terms already detect  $\text{HOD}$  [\[04\]](#)). *The equivalent conditions in [theorem 7.2](#) remain equivalent if condition (iv) is replaced by “every member of  $\mathcal{U}_2$  is ordinal definable.”*

*Proof.* Use the two mutually definable maps in [proposition 6.5](#) to recover every surreal from such a two-term omnific integer, and then apply [theorem 7.2](#).  $\square$

*Remark 7.4.* There is no Choice-based definable selection hidden in the converse. We choose a code only to prove its existence; hypothesis (ii) says *every* such surreal code is ordinal definable. This is much weaker as a proof requirement than constructing a canonical code from an arbitrary set in a universe not satisfying  $V = \text{HOD}$ .

**Theorem 7.5** (Hamkins, 2012; included for comparison [\[07\]](#)). *The ordinal-definable surreals are dense in  $\mathbf{No}$  if and only if  $V = \text{HOD}$ . Hamkins further proves equivalence with a parameter-free definable map  $\text{Ord} \rightarrow \mathbf{No}$  having dense image, and with definable surjectivity and bijectivity in that setting [\[13\]](#).*

*Proof of the density equivalence.* If  $x$  is a sign sequence which is not OD, let  $x^-$  and  $x^+$  be its extensions by one minus sign and one plus sign. Every number strictly between  $x^-$  and  $x^+$  extends  $x$ : an earlier disagreement, or termination before the length of  $x$ , puts it outside this interval. If such a number  $y$  were OD, then the prefix  $y \upharpoonright b(x)$  would be OD using the ordinal parameter  $b(x)$ . That prefix is  $x$ , a contradiction. Thus density forces every surreal to be OD, and [theorem 7.2](#) yields  $V = \text{HOD}$ . The converse is immediate.  $\square$

The proof just given is the prefix argument from Hamkins’s earlier result, not a new density proof claimed here. The bounded tests use the explicit omnific codes, whereas the density criterion is already known.

**Proposition 7.6** (Specialization of the old-field theorem [07]). *If  $V \neq \text{HOD}$ , the subclass  $\mathbf{No}^{\text{HOD}}$  is closed and nowhere dense in the fine order topology of  $\mathbf{No}$ .*

*Proof.* For every  $x \notin \mathbf{No}^{\text{HOD}}$ , the interval  $(x^-, x^+)$  in the previous proof contains no HOD surreal. Hence the complement is open, and  $\mathbf{No}^{\text{HOD}}$  is closed.

For completeness, any proper real closed subfield  $F \subsetneq \mathbf{No}$  has empty interior. Suppose an open interval were contained in  $F$ , and choose  $c < d$  in that interval. For arbitrary  $y \in \mathbf{No}$ ,  $z = c + (d - c)\tau(y)$  lies between  $c$  and  $d$ , so  $z \in F$ . Then  $(z - c)/(d - c) \in F$ , and the algebraic inverse (6.4), using the positive square root in the real closed field  $F$ , implies  $y \in F$ . This would give  $F = \mathbf{No}$ . Apply this to the proper field  $\mathbf{No}^{\text{HOD}}$  and combine empty interior with closedness.  $\square$

The universes report proves the more general assertion for proper same-ordinal inner models, including its surcomplex analogue (`univ:thm:nowheredense`; source 07 cited it as “Theorem 14.3” of that report). [Proposition 7.6](#) is the case  $M = \text{HOD}$  of that theorem and is not counted as a new result. It also illustrates the difference between fine surreal topology and the usual topology on the embedded real line.

**Theorem 7.7** (The pointwise-definability test [04, 07]; items (v)–(vi) for arbitrary set models [merge]). *Let  $M \models \text{ZFC}$  be any set model. In external semantics, the following are equivalent:*

- (i)  $M$  is pointwise definable;
- (ii) every member of  $\mathbf{No}^M$  is parameter-free definable in  $M$ ;
- (iii) every member of  $\mathbf{Oz}^M$  is parameter-free definable in  $M$ ;
- (iv) every member of  $\mathcal{U}^M$  is parameter-free definable in  $M$  [04];
- (v) every omnific integer  $z$  of  $M$  with  $0 < z < \omega$  is parameter-free definable in  $M$  [07];
- (vi) every monomial  $\omega^a$  of  $M$  with  $0 < a < 1$  is parameter-free definable in  $M$  [07].

*Proof.* The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) and (iii) $\Rightarrow$ (v) $\Rightarrow$ (vi) are immediate. Each  $s \in \mathbf{No}^M$  is recovered by a fixed standard formula from  $\mathcal{C}^M(s) \in \mathcal{U}^M$ , and each  $x \in \mathbf{No}^M$  from the monomial  $\mathbf{Q}^M(x)$ , so (iv) and (vi) each imply (ii). Internally to  $M$ , every set has a code as in [lemma 7.1](#). For any fixed  $a \in M$ , choose such a code  $s \in M$ . A standard formula defines  $s$  by (ii), and composing it with the standard decoder formula defines  $a$ . This does not require  $M$ ’s internal relation to be externally well-founded: the collapse operation is the one that  $M$  proves exists uniquely.  $\square$

Source 04 proved (i)–(iv) for arbitrary set models; source 07 proved (i)–(iii), (v) and (vi) for transitive  $M$ . The argument for (vi) $\Rightarrow$ (ii) uses only internal formulas, so 04’s proof covers 07’s conditions for arbitrary set models; this extension is the merge’s.

Pointwise definable models of ZFC exist whenever ZFC is consistent; Hamkins–Linetsky–Reitz also prove strong extension theorems and treat nonstandard as well as transitive models [12]. Their existence theorem is prior work, and no existence assumption is needed for the equivalence. The new feature of [theorem 7.7](#) is the restriction of the test to a single fixed-leading-term or bounded omnific interval. It also explains why an external counting argument cannot be converted into an internal proof that some member of  $\mathbf{Oz}^M$  is undefinable: the formulas that define particular elements are not presented to  $M$  as a surjective internal list. In a pointwise definable model every set can be definable while the model still satisfies its own uncountability assertions.

## 8 From one missing number to proper-class transcendence

This section contains the strongest algebraic result, due to source 04. We first isolate a support argument that does not involve definability at all.

**Lemma 8.1** (Disjoint support cosets [04]). *Let  $K \subseteq \mathbf{No}$  be a subfield, and let  $G \subseteq (\mathbf{No}, +)$  be an additive subgroup such that*

$$\text{supp}(a) \subseteq G \quad (0 \neq a \in K).$$

*Suppose  $(g_i)_{i \in J}$  is a family such that every nonzero finitely supported integer combination satisfies*

$$\sum_i m_i g_i \notin G. \quad (8.1)$$

*Then the family  $(\omega^{g_i})_{i \in J}$  is algebraically independent over  $K$ . The assertion applies to a class family when the condition is read on each finite subfamily.*

*Proof.* Take a nonzero polynomial in finitely many distinct variables,

$$P(Y_1, \dots, Y_n) = \sum_{\nu \in F} a_\nu Y_1^{\nu_1} \cdots Y_n^{\nu_n},$$

where  $F \subseteq \mathbb{N}^n$  is finite and the displayed coefficients are nonzero. Since  $\omega^u \omega^v = \omega^{u+v}$ , the normal-form support of the term at multi-index  $\nu$  after substitution is

$$\text{supp}(a_\nu) + \sum_{j=1}^n \nu_j g_j \subseteq G + \sum_{j=1}^n \nu_j g_j. \quad (8.2)$$

For  $\nu \neq \mu$ , these cosets are disjoint. Otherwise subtracting an exponent in their intersection would give  $\sum_j (\nu_j - \mu_j) g_j \in G$ , contrary to (8.1). Every displayed term is nonzero, and its coefficients cannot be cancelled by any other term. The finite sum is therefore nonzero. This proves the required absence of polynomial relations. No infinite polynomial, class sum, or interchange of summations is used.  $\square$

**Theorem 8.2** (Support-subfield amplification [04]). *Let  $K$  be a proper subfield of  $\mathbf{No}$  satisfying*

$$\omega \in K, \quad e_\alpha \in K \ (\alpha \in \text{Ord}), \quad \text{supp}(a) \subseteq K \ (0 \neq a \in K), \quad (8.3)$$

*where  $e_\alpha = \omega^{-(\alpha+1)}$ . For any  $t \in (0, 1) \setminus K$ , the class family*

$$y_\alpha = \omega^{te_\alpha}, \quad z_\alpha = \omega + y_\alpha \quad (\alpha \in \text{Ord}) \quad (8.4)$$

*is algebraically independent over  $K$ , both for the  $y$  family and for the  $z$  family. Each  $z_\alpha$  lies in  $\mathcal{U}$  and has exactly two nonzero normal-form terms, both with coefficient 1. Such a  $t$  exists for every proper  $K$  satisfying (8.3).*

*Proof.* First, the family  $(e_\alpha)$  is linearly independent over  $\mathbb{Q}$ . Indeed, a finite rational linear combination is already a normal form with distinct exponents  $-(\alpha + 1)$ ; after deleting zero coefficients it cannot vanish. Hence for distinct  $\alpha_1, \dots, \alpha_n$  and integers  $m_j$  not all zero,

$$c = \sum_{j=1}^n m_j e_{\alpha_j} \in K \setminus \{0\}.$$

If  $\sum_j m_j t e_{\alpha_j} = t c$  belonged to  $K$ , division by  $c$  would give  $t \in K$ , a contradiction. Thus the exponents  $t e_\alpha$  satisfy (8.1) for the additive subgroup  $G = (K, +)$ . Lemma 8.1 proves algebraic independence of the  $y_\alpha$ .

Translation by  $\omega \in K$  is a polynomial automorphism on each finite tuple of variables. Therefore a nonzero relation among the  $z_\alpha = \omega + y_\alpha$  would give a nonzero relation among the  $y_\alpha$ , which is impossible.

Finally,

$$0 < t e_\alpha < e_\alpha \leq \omega^{-1} < 1/2.$$

Consequently  $\omega^{t e_\alpha}$  is a positive, purely infinite omnific monomial strictly below  $\omega^{1/2}$ . The exponents 1 and  $t e_\alpha$  are distinct and positive, so the asserted two-term normal form is exact and  $z_\alpha \in \mathcal{U}$ .

For existence, take  $x \in \mathbf{No} \setminus K$ . Then  $u = |x| > 0$  is not in  $K$ , since otherwise  $x = u$  or  $x = -u$  would be in  $K$ . Set  $t = u/(1 + u)$ . It lies strictly between 0 and 1. If  $t \in K$ , then  $u = t/(1 - t) \in K$ , again a contradiction.  $\square$

**Theorem 8.3** (The HOD-surreal transcendence dichotomy [04]). *Exactly one of the following occurs:*

- (i)  $V = \text{HOD}$  and  $\mathbf{No} = K_{\text{HOD}}$ ;
- (ii)  $V \neq \text{HOD}$ , and for any  $t \in (0, 1) \setminus K_{\text{HOD}}$ , the family

$$\left( \omega + \omega^{t \omega^{-(\alpha+1)}} \right)_{\alpha \in \text{Ord}}$$

*is algebraically independent over  $K_{\text{HOD}}$  and is contained in  $\mathcal{U}$ .*

*In case (ii), for every cardinal  $\kappa$  there is an algebraically independent subset of  $\mathcal{U}$  of cardinality  $\kappa$  over  $K_{\text{HOD}}$ . In particular, the transcendence defect has no set-sized cardinal bound.*

*Proof.* Theorem 7.2 identifies  $V \neq \text{HOD}$  with  $K_{\text{HOD}} \neq \mathbf{No}$ . The field  $K_{\text{HOD}}$  contains  $\omega$  and every  $e_\alpha$  because these are ordinal definable. Its support property is lemma 5.2. Apply theorem 8.2. Restricting the class family to  $\alpha < \kappa$  gives a set family by Replacement. Algebraic independence implies distinctness, hence that set has cardinality  $\kappa$ .  $\square$

*Remark 8.4* (What “proper-class transcendence” means [04]). The theorem gives a concrete class injection with finite polynomial independence, definable from one parameter  $t$ . It does not choose a transcendence basis for  $\mathbf{No}$ , invoke Global Choice, or assign an ordinary cardinal to a proper class. Each individual  $z_\alpha$  is transcendental over  $K_{\text{HOD}}$ , but this pointwise assertion alone would be much weaker than the simultaneous algebraic independence proved above. In a set model, the assertions have their internal cardinal and ordinal meaning, not an assertion that its externally set-sized carrier is literally a proper class in the metatheory.

**Corollary 8.5** (Localization at other definable scales [04]). *Assume  $V \neq \text{HOD}$ . For every  $p \in I_{\text{HOD}}$ , every positive  $\gamma \in K_{\text{HOD}}$ , and every  $t \in (0, 1) \setminus K_{\text{HOD}}$ , the family*

$$w_\alpha = p + \omega^{\gamma t e_\alpha} \quad (\alpha \in \text{Ord}) \tag{8.5}$$

*is algebraically independent over  $K_{\text{HOD}}$  and lies in  $(p, p + \omega^\gamma) \cap \mathbf{Oz}$ .*

*Proof.* For every nonzero finite integer combination  $c$  of the  $e_\alpha$ , we have  $c \in K_{\text{HOD}} \setminus \{0\}$ . If  $\gamma tc \in K_{\text{HOD}}$ , division by  $\gamma c$  would put  $t$  in  $K_{\text{HOD}}$ . The coset proof therefore applies to the exponents  $\gamma te_\alpha$ . Translation by  $p \in K_{\text{HOD}}$  preserves independence. The exponents are positive and less than  $\gamma$ , so the added monomials are omnific and lie strictly between 0 and  $\omega^\gamma$ .  $\square$

The widths in this corollary are infinite monomial widths. It would be false to replace them indiscriminately by positive real widths: an interval of width less than 1 contains at most one omnific integer. The result is localization at an arbitrary definable *infinite scale*, not density of omnific integers in **No**.

## 9 Uniform definability is not termwise definability

There are two different pitfalls. Individually definable terms need not form a definable sequence; conversely, a definable sequence need not have definable entries at arbitrary ordinal indices. Each source gives an explicit example of each; both pairs are kept, since the mechanisms differ.

**Theorem 9.1** (Definable support and terms do not suffice [04]). *Assume  $V \neq \text{HOD}$ . There is a valid normal form  $q$  satisfying all of the following:*

- (i)  $q \in \mathcal{U} \setminus I_{\text{HOD}}$ ;
- (ii) *its support, as a whole set, belongs to  $\text{HOD}$ ;*
- (iii) *every coefficient is in  $\{1, 2, 3\}$ , and every individual monomial term belongs to  $K_{\text{HOD}}$ ;*
- (iv) *the whole indexed coefficient/exponent sequence does not belong to  $\text{HOD}$ .*

*Thus  $K_{\text{HOD}}$  is not closed under arbitrary ambient Hahn-summable families of its elements, even when the support belongs to  $\text{HOD}$ .*

*Proof.* Choose  $s \notin K_{\text{HOD}}$  by [theorem 7.2](#), and let  $\alpha = b(s)$  and  $q = C(s)$ . The reversible code gives  $q \notin I_{\text{HOD}}$ . Its support is the set

$$\{1\} \cup \{e_\beta : \beta \leq \alpha\},$$

which is definable from the ordinal  $\alpha$  and whose members all belong to  $\text{HOD}$ . Hence the support itself belongs to  $\text{HOD}$ . Every individual coefficient is one of three fixed rationals, and every individual exponent is ordinal definable, so every individual term is ordinal definable. If the whole normal form belonged to  $\text{HOD}$ , [lemma 5.2](#) would give  $q \in K_{\text{HOD}}$ , a contradiction. The summable family here is simply the family of its distinct monomial terms, so no additional summability issue occurs.  $\square$

*Remark 9.2.* In this example even the *set of coefficient values* is a finite  $\text{HOD}$  set. What fails is the definability of which coefficient is assigned to which exponent. Saying “the coefficients are rational” erases precisely the information carrying the original sign sequence.

**Theorem 9.3** (Uniform subset coding by omnific normal forms [07]). *For an ordinal  $\alpha$  and  $X \subseteq \alpha$ , put  $\iota_\beta = 1/(\underline{\beta} + 2)$  and*

$$E_\alpha(X) = \omega^{3/4} + \sum_{\beta < \alpha} (1 + \chi_X(\beta)) \omega^{\iota_\beta}. \quad (9.1)$$

*Then this is a well-defined omnific integer satisfying*

$$\omega^{3/4} \leq E_\alpha(X) < 2\omega^{3/4} < \omega, \quad (9.2)$$

with strictness in the first inequality when  $\alpha > 0$ . The pair  $(\alpha, X)$  and  $E_\alpha(X)$  are mutually ambient-definable without additional parameters. In particular,

$$E_\alpha(X) \in K_A^M \iff \alpha, X \in D_A^M,$$

and the analogous equivalence holds with ordinal definability in place of  $A$ -definability.

*Proof.* The ordinal embedding is strictly increasing, so the exponents  $\iota_\beta$  are strictly decreasing and positive. Their largest possible value is  $1/2$ , below the marker exponent  $3/4$ . Thus (9.1) is already a decreasing normal form, all of whose exponents are positive. It is omnific. Its leading term has coefficient 1 at exponent  $3/4$ , so its lower terms are strictly smaller in Archimedean order; this proves (9.2).

Starting from  $(\alpha, X)$ , Replacement and the normal-form construction define the number uniquely. Starting from the number, take its canonical normal form, remove the first, marker term, and take the order type of the remaining decreasing support. This is  $\alpha$ , including the empty case. The coefficient at the  $\beta$ th remaining place is 2 precisely when  $\beta \in X$ . Hence  $X$  is recovered uniquely. The definability equivalences follow from composition.  $\square$

The marker in (9.1) is necessary for this simple uniform empty-case decoder and bound. No cancellation can obscure a bit: the exponents are distinct and the coefficients are never zero. The bit code and 04's sign code  $C$  exploit the same phenomenon;  $C$  is self-delimiting through its coefficient alphabet  $\{1, 2, 3\}$ , while  $E_\alpha$  uses a marker term and lands below  $\omega$ .

**Corollary 9.4** (Individually definable entries are insufficient [07]). *Suppose  $X \subseteq \omega$  belongs to  $M$  but is not  $A$ -definable. Then  $E_\omega(X)$  is not  $A$ -definable, although its support is parameter-free definable and every one of its real coefficients and exponents is individually parameter-free definable. If  $X$  is not ordinal-definable, the entire number is not ordinal-definable either.*

*Proof.* The support is the fixed set  $\{3/4\} \cup \{1/(n+2) : n \in \mathbb{N}\}$ . Its entries are rational, and each coefficient is either 1 or 2. Nondefinability of the sum follows from the decoder in theorem 9.3.  $\square$

There is no contradiction in saying that every coefficient is definable while the coefficient assignment is not. The statement “the coefficient at  $n$  is 1” or “it is 2” may require a nonuniform choice of which true statement to use. The missing information is exactly the subset  $X$ . For arbitrary  $X \subseteq \alpha$ , all exponents in (9.1) are ordinal-definable and the support is definable from  $\alpha$ ; a non-OD  $X$  still gives a non-OD sum. Thus even allowing ordinal parameters does not turn pointwise definability of data into uniform definability.

**Proposition 9.5** (A definable integer can encode nondefinable entries [04]). *There is a parameter-free ambiently definable  $q_* \in \mathcal{U}$  whose normal-form coefficients canonically encode every member of  $\mathbb{R} \cap \text{HOD}$ . In any universe or fixed transitive model in which some member of  $\mathbb{R} \cap \text{HOD}$  is not parameter-free definable,  $q_*$  has a normal-form coefficient that is not parameter-free definable. The last hypothesis is essential and is not asserted in every model.*

*Proof.* The set  $R_H = \mathbb{R} \cap \text{HOD}$  is parameter-free definable. Restrict the canonical set-like HOD well-order to it, and let  $(r_\xi)_{\xi < \lambda}$  be the unique increasing enumeration. The entire enumeration and its ordinal order type  $\lambda$  are parameter-free definable. Define the injective real function

$$\zeta(r) = \frac{3}{2} + \frac{r}{2(1+|r|)} \in (1, 2).$$

Its inverse on  $(1, 2)$  is parameter-free definable: if  $u = 2\zeta(r) - 3$ , then  $r = u/(1 - |u|)$ . Form

$$q_* = \omega + \sum_{\xi < \lambda} \zeta(r_\xi) \omega^{e_\xi}. \quad (9.3)$$

This is a valid normal form and is parameter-free definable from the whole enumeration. The set  $R_H$  is nonempty, and the leading tail exponent is  $e_0 < 1/2$ , so  $q_* \in \mathcal{U}$ . Every  $r \in R_H$  occurs in exactly one coefficient, after applying  $\zeta$ . If that  $r$  is not parameter-free definable, neither is  $\zeta(r)$ , since the displayed inverse would otherwise define  $r$ .  $\square$

This does not contradict unique-output closure. Extracting the coefficient at position  $\xi$  uses  $\xi$  as an additional parameter; a parameter-free definition of  $q_*$  need not define every such ordinal position. Ordinal definability allows that parameter, which is why all coefficients of an ordinal-definable surreal do belong to HOD.

**Theorem 9.6** (The reverse uniformity gap [07]). *Suppose  $\mu \in \text{Ord}^M$  is  $A$ -definable and there is  $\beta < \mu$  which is not  $A$ -definable. Then*

$$U_\mu = \sum_{\gamma < \mu} \omega^{\iota_\gamma}, \quad \iota_\gamma = \frac{1}{\underline{\gamma} + 2}, \quad (9.4)$$

*belongs to  $I_A^M \cap (0, \omega)$ , but its exponent  $\iota_\beta$  and its corresponding monomial are not  $A$ -definable.*

*Proof.* The entire normal-form family in (9.4) is uniformly defined from  $\mu$ . Hence so is its sum, which is a positive omnific integer with leading exponent  $1/2$ . If  $\iota_\beta$  were  $A$ -definable, taking its reciprocal and subtracting 2 would make the surreal ordinal  $\underline{\beta}$ , and therefore the set-theoretic ordinal  $\beta$ ,  $A$ -definable. If the monomial were  $A$ -definable, applying  $\Omega^{-1}$  would give the same contradiction.  $\square$

This theorem is conditional on the stated ordinal configuration. One must not assume that an ordinal which is uncountable *inside* a countable transitive model necessarily has an externally undefinable predecessor; pointwise definable models are a reminder that it need not. Section 10 gives an exact criterion for when the reverse gap can occur. The collector (04) obtains a nondefinable coefficient from real information in HOD; the reverse gap (07) obtains a nondefinable exponent from ordinal information. The two are different mechanisms.

**A compact lesson for proposed definitions [07].** Defining a “definable surreal” to mean “a normal form with definable coefficients and exponents” produces an unstable notion unless a representation and a uniformity condition are specified. Theorems 9.1 and 9.3 show that it can admit numbers with an undefinable information sequence. Proposition 9.5 and theorem 9.6 show that it can reject uniformly definable numbers. Using the entire canonical graph, as in proposition 4.9, avoids both errors and agrees exactly with ambient definability.

## 10 The maximal initial definable core

Throughout this section, which is source 07’s,  $M$  is an *externally fixed transitive* set model of ZFC and  $A \subseteq M$ ; the proofs use a least externally omitted ordinal, which need not exist for a nonstandard model. We write  $K = K_A^M$ ,  $I = I_A^M$  and  $O = O_A^M = \text{Ord}^M \cap D_A^M$ . The next elementary observation links the simplicity tree to definability of ordinals.

**Proposition 10.1** (Initiality criterion [07]). *The subclass  $K_A^M$  is initial in  $\text{No}^M$  if and only if  $O_A^M$  is downward closed in  $\text{Ord}^M$ .*

*Proof.* If  $K_A^M$  is initial and  $\alpha \in O_A^M$ , the all-plus surreal  $\underline{\alpha}$  is in  $K_A^M$ . Each  $\beta$  for  $\beta < \alpha$  is a prefix, hence definable. Its birthday is  $\beta$ , so  $\beta \in O_A^M$ . Conversely, let  $x \in K_A^M$ . Its birthday  $\alpha$  is definable from  $x$ , so  $\alpha \in O_A^M$ . If  $\beta < \alpha$ , downward closure makes  $\beta$   $A$ -definable. The prefix  $x \upharpoonright \beta$  is uniquely definable from  $x, \beta$  and therefore belongs to  $K_A^M$ .  $\square$

## 10.1 The first omitted ordinal is an epsilon number

Assume  $O_A^M \neq \text{Ord}^M$ . Since  $M$  is transitive, the external nonempty set  $\text{Ord}^M \setminus O_A^M$  has a least element. Denote it by

$$\rho_A^M = \min(\text{Ord}^M \setminus O_A^M). \quad (10.1)$$

This is a metatheoretic definition. It does not say that the minimum is defined from  $A$  in  $M$ .

**Lemma 10.2** (Cantor-normal-form closure [07]). *The ordinal  $\rho = \rho_A^M$  satisfies  $\omega^\rho = \rho$ , where the power is ordinal exponentiation. In particular  $\rho > \omega$ .*

*Proof.* The ordinals 0 and all finite ordinals are parameter-free definable. Suppose a nonzero  $\rho$  were not a fixed point of ordinal exponentiation with base  $\omega$ . Its finite Cantor normal form would be

$$\rho = \omega^{\gamma_0} n_0 + \cdots + \omega^{\gamma_{k-1}} n_{k-1}, \quad \gamma_0 > \cdots > \gamma_{k-1}, \quad n_j \in \mathbb{N}_{>0},$$

with every  $\gamma_j < \rho$ . Indeed, the leading exponent is at most  $\rho$ ; equality would force  $\rho = \omega^\rho$ , and would leave no further term or coefficient greater than one. All exponents in this finite expression are  $A$ -definable by minimality of  $\rho$ . Ordinal exponentiation, finite ordinal sums, and finite coefficients are uniquely definable operations. The finite expression would therefore make  $\rho$   $A$ -definable, a contradiction. The least epsilon number is greater than  $\omega$  (and is itself parameter-free definable), so the final assertion follows.  $\square$

The argument is ordinary ordinal definability, not a new theorem about Cantor normal form. Its role is to identify exactly the cutoff at which the classical epsilon-field theorem becomes available.

## 10.2 The core theorem

**Theorem 10.3** (Maximal initial elementary exponential core [07]). *Let  $M$  be transitive, and suppose  $\rho = \rho_A^M$  exists. Define*

$$\text{Core}_A^M = K_A^M \cap (\mathbf{No}_{<\rho})^M. \quad (10.2)$$

*Then:*

- (i)  $\text{Core}_A^M$  is initial, and every initial subclass of  $K_A^M$  is contained in it.
- (ii) Its surreal ordinals are exactly the ordinals below  $\rho$ .
- (iii) It is an elementary exponential subfield of  $\mathbf{No}^M$ , hence real closed and closed under positive logarithms.
- (iv)  $\text{Core}_A^M \cap \mathbf{Oz}^M$  is an integer part of  $\text{Core}_A^M$ .
- (v)  $\text{Core}_A^M \cap \mathbb{R}^M = K_A^M \cap \mathbb{R}^M$ .

*If  $O_A^M = \text{Ord}^M$ , the same initiality and elementary-field conclusions hold with  $\text{Core}_A^M = K_A^M$ .*

*Proof.* Let  $x \in \text{Core}_A^M$ , and put  $\alpha = \text{b}(x) < \rho$ . Every  $\beta < \alpha$  is  $A$ -definable by the definition of  $\rho$ . Hence  $x \restriction \beta$  is  $A$ -definable and has birthday below  $\rho$ . This proves initiality.

No  $A$ -definable surreal has birthday exactly  $\rho$ , since the birthday is definable from the number. If an initial subclass  $S \subseteq K_A^M$  contained  $x$  with birthday greater than  $\rho$ , it would also contain  $x \restriction \rho$ . The birthday of that prefix is  $\rho$ , giving the same contradiction. Therefore every such  $S$  is contained in (10.2). This proves (i), and (ii) follows because every ordinal below  $\rho$  is definable and the all-plus sequence has that ordinal as birthday.

By lemma 10.2,  $\rho$  is an epsilon number. Thus the classical cutoff theorem (section 2.5) gives

$$(\mathbf{No}_{<\rho})^M_{\text{exp}} \prec \mathbf{No}^M_{\text{exp}}.$$

Theorem 4.4 gives  $(K^M_A)_{\text{exp}} \prec \mathbf{No}^M_{\text{exp}}$ . Fix the parameter-free definable selection functions of lemma 4.3 for the common ambient exponential field. Each elementary substructure is closed under them: for parameters in it, the uniquely defined selected value belongs to it by elementarity. Their intersection is therefore closed under the same selectors. By lemma 4.3, the intersection in (10.2) is elementary. This proves (iii). This step uses common definable selectors; it is not an unsupported assertion that arbitrary intersections of elementary submodels are elementary.

For (iv), the omnific floor of  $x \in \text{Core}^M_A$  belongs to  $K^M_A$  by theorem 4.1 and is a prefix of  $x$  by proposition 2.3. Initiality therefore places the floor in  $\text{Core}^M_A$ . Finally every embedded real has birthday at most  $\omega < \rho$ , giving (v).

If all ordinals of  $M$  are  $A$ -definable, proposition 10.1 makes  $K^M_A$  initial, and theorem 4.4 gives the remaining field assertion.  $\square$

**Corollary 10.4** (Parameter monotonicity [07]). *If  $A \subseteq A' \subseteq M$ , then  $\text{Core}^M_A \subseteq \text{Core}^M_{A'}$ , interpreting the core as in theorem 10.3 in the no-omission case. When both first omitted ordinals exist,  $\rho^M_A \leq \rho^M_{A'}$ .*

*Proof.* The initial class  $\text{Core}^M_A$  is contained in  $K^M_{A'}$ , so the maximality of  $\text{Core}^M_{A'}$  gives the inclusion. Every ordinal below  $\rho^M_{A'}$  is also  $A'$ -definable, giving the ordinal inequality.  $\square$

**Theorem 10.5** (Birthday-controlled omnific denominators [07]). *Let  $\epsilon \in \text{Ord}^M$  be any epsilon number, and put*

$$K_{A,<\epsilon} = K^M_A \cap (\mathbf{No}_{<\epsilon})^M, \quad I_{A,<\epsilon} = I^M_A \cap (\mathbf{No}_{<\epsilon})^M.$$

*Then  $\text{Frac}(I_{A,<\epsilon}) = K_{A,<\epsilon}$ . For nonzero  $x \in K_{A,<\epsilon}$  with  $\alpha = \text{b}(x)$ , the explicit choice*

$$q = \omega^{\alpha+1}, \quad p = qx \tag{10.3}$$

*gives  $p, q \in \Pi^M \cap K_{A,<\epsilon}$  and  $x = p/q$ . The rule is definable from  $x$  alone, without  $\epsilon$  as an additional parameter. In particular,*

$$\text{Frac}(\text{Core}^M_A \cap \mathbf{Oz}^M) = \text{Core}^M_A.$$

*Proof.* For each exponent  $a$  of  $x$ , lemma 2.6 gives  $\text{b}(a) \leq \alpha$ , hence  $a \geq -\underline{\alpha}$ . Therefore

$$a + \underline{\alpha + 1} \geq -\underline{\alpha} + (\underline{\alpha} + 1) = 1 > 0.$$

All exponents of  $p$  are positive, so  $p$  and  $q$  are purely infinite. The number  $q$  is the surreal ordinal corresponding to the ordinal  $\omega^{\alpha+1}$ . As  $\epsilon$  is an epsilon number and  $\alpha < \epsilon$ , we have  $\alpha + 1 < \epsilon$  and  $\omega^{\alpha+1} < \epsilon$ . Thus  $\text{b}(q) < \epsilon$ . The epsilon-cutoff field is closed under multiplication, so  $p = qx$  also has birthday below  $\epsilon$ . Both  $p$  and  $q$  are definable from  $x$ , and hence belong to  $K^M_A$ . The zero case is  $0/1$ . The core assertion follows by taking  $\epsilon = \rho^M_A$  when that ordinal exists, and by theorem 4.1 in the no-omission case.  $\square$

**Corollary 10.6** (Normal-form heredity and valuation of the core [07]). *The field  $\text{Core} = \text{Core}^M_A$  is closed under  $\Omega$  and its inverse on monomials. Every exponent, coefficient, and monomial term in the normal form of an element of  $\text{Core}$  belongs to  $\text{Core}$ . Its normal-form value group is  $(\text{Core}, +)$ , and its residue field is  $R_A = K^M_A \cap \mathbb{R}^M$ .*

*Proof.* Suppose first that  $\rho = \rho_A^M$  exists. The bound  $b(\Omega(a)) \leq \omega^{b(a)} < \rho$  for  $a \in \text{Core}$ , together with ambient definability, proves closure under  $\Omega$ . The opposite birthday inequality gives closure under its inverse on monomials.

If  $x \in \text{Core}$ , its normal-form length is at most  $b(x) < \rho$ . Every index in that length is therefore  $A$ -definable. [Proposition 4.9](#) makes each coefficient and exponent definable. Their birthdays, and those of the corresponding terms, are below  $\rho$  by [lemma 2.6](#); coefficients are real and hence have birthday at most  $\omega$ . Thus they lie in  $\text{Core}$ . In the no-omission case, every ordinal index is definable and the same reasoning requires no birthday cutoff. Now repeat the proof of [theorem 4.10](#): leading exponents stay in  $\text{Core}$ ,  $\omega^{-a} \in \text{Core}$  realizes every value  $a \in \text{Core}$ , and standard part stays in the common real subfield from [theorem 10.3\(v\)](#).  $\square$

### 10.3 What the core theorem does and does not say

The full ambient-definable field need not be initial: a definable high ordinal can have an undefinable predecessor. Nevertheless it always has the canonical initial elementary field just constructed. The reverse normal-form gap of [theorem 9.6](#) occurs exactly when  $O_A^M$  is not downward closed, equivalently when  $K_A^M$  is not initial.

The ordinal  $\rho$  is not an additional allowed parameter in the definition of  $K_A^M$ . It indexes an externally described intersection. Calling it “the first undefinable ordinal” in the metatheory does not turn it into a definition inside  $M$ . This is precisely the distinction that prevents a Berry-style contradiction.

The fraction-field conclusion required the additional birthday argument of [theorem 10.5](#); it was not automatic from initiality or real closedness. In general an initial real closed subfield need not be the fraction field of its omnific integer part: the real algebraic subfield has omnific integer part  $\mathbb{Z}$ . Here it is the epsilon cutoff and the definability of the ordinal denominator that remove this obstruction.

The ordinal  $\rho_A^M$  and 04’s threshold  $\lambda_H$  ([section 11](#)) are different invariants:  $\rho$  is the first ordinal not definable from  $A$ , and  $\lambda_H$  is the first birthday of a non-ordinal-definable surreal. With ordinal parameters every ordinal is definable, so the core of the ordinal-definable field is the whole field  $K_{\text{HOD}}$ , which is initial ([theorem 5.1](#)).

**Proposition 10.7** (Failure of finite-parameter continuity [07]). *Suppose  $M$  contains an ordinal  $\mu$  which is uncountable in the external metatheory. Let  $A = \{\beta : \beta \leq \mu\} \subseteq M$ . Then*

$$\bigcup_{A_0 \subseteq A \text{ finite}} \text{Core}_{A_0}^M \subsetneq \text{Core}_A^M.$$

*Proof.* For finite  $A_0$ , only countably many ordinals are externally  $A_0$ -definable. Some ordinal below  $\mu$  is therefore omitted, so  $\rho_{A_0}^M < \mu$ . The surreal ordinal  $\underline{\mu}$  is not in  $\text{Core}_{A_0}^M$ , since its birthday exceeds that cutoff. On the other hand every ordinal at most  $\mu$  is itself an allowed parameter in  $A$ . Thus the first omitted ordinal for  $A$ , if there is one, is greater than  $\mu$ , and  $\underline{\mu} \in \text{Core}_A^M$ . Parameter monotonicity gives the inclusion, and this element proves it strict.  $\square$

This is a conditional statement about externally uncountable transitive models. It does not apply merely because  $M$  calls an ordinal uncountable. It shows that, although each individual definition uses finitely many parameters, requiring *all* prefixes to remain definable is a genuinely non-finitary condition.

## 11 The first birthday of a non-ordinal-definable surreal

When  $V \neq \text{HOD}$ , there is a genuine definable first place where ordinal-definable signs cease to exhaust the construction. It is not the birthday of a canonically selected nondefinable number.

**Definition 11.1** ([04]). Assuming  $V \neq \text{HOD}$ , define

$$\lambda_H = \min\{\alpha \in \text{Ord} : \mathcal{P}(\alpha) \not\subseteq \text{HOD}\}. \quad (11.1)$$

Existence follows from the set-of-ordinals coding in [lemma 7.1](#). All terms in this definition are first-order definable classes or set constructions.

$\text{HOD}$  is a parameter-free definable transitive inner class of  $V$  containing every ordinal and satisfying ZFC, so the pair  $M = \text{HOD}$ ,  $N = V$  satisfies the standing hypothesis (H) of the universes report, and  $\lambda_H = \beta(\text{HOD}, V)$  in that report's notation.

**Theorem 11.2** (Exact birthday threshold [04]; the case  $M = \text{HOD}$  of `univ:prop:beta`). *Assume  $V \neq \text{HOD}$ . Then*

$$\lambda_H = \min\{b(s) : s \in \mathbf{No} \setminus K_{\text{HOD}}\}. \quad (11.2)$$

*The ordinal  $\lambda_H$  is an infinite cardinal in both  $\text{HOD}$  and  $V$ . It is itself a parameter-free definable ordinal, and therefore a parameter-free definable omnific integer.*

*Proof.* The equality (11.2) and the two cardinality assertions are `univ:prop:beta` of the universes report [21] (from that report's source 08), applied with  $M = \text{HOD}$  and  $N = V$  and combined with  $K_{\text{HOD}} = \mathbf{No}^{\text{HOD}}$  ([theorem 5.1](#)); that report also proves  $\delta(\text{HOD}, V) \leq \text{cf}(\lambda_H)$  for its omitted-cut invariant  $\delta$ . Its proof is the argument source 04 gives: signs of length  $\alpha$  and subsets of  $\alpha$  are interdefinable from  $\alpha$ ; a  $\text{HOD}$  bijection with a smaller ordinal would pull a new subset back; and a  $V$ -bijection with a smaller cardinal would give a  $\text{HOD}$  code of a well-order of type  $\lambda_H$ . The birthday-cutoff report proves the equality and the inner-model cardinality for transitive models  $M \subsetneq N$  of ZFC with the same ordinals as `hset:thm:firstbirthday`, which covers  $\text{HOD}^N \subseteq N$  for a transitive set model  $N$ . The remaining assertion is 04's: (11.1) is a parameter-free unique definition of  $\lambda_H$ , since  $\text{HOD}$  is parameter-free definable, and ordinal surreals are omnific.  $\square$

**Corollary 11.3** (No definable witness selection [04]). *The set of non-ordinal-definable signs of length  $\lambda_H$  is a nonempty parameter-free definable set, but none of its members is ordinal definable. In particular, there is no ordinal-definable selection of one of those signs, or of one of their omnific codes.*

*Proof.* The set is defined using the definable predicate  $\text{HOD}$  and the parameter-free ordinal  $\lambda_H$ . Its nonemptiness follows from the definition of that ordinal. A selected ordinal-definable member would contradict the defining condition. An ordinal-definable code would yield such a member by the decoder.  $\square$

### 11.1 A current set-theoretic restriction on the threshold

The threshold is the first cardinal where  $V$  and  $\text{HOD}$  disagree about subsets. It therefore inherits restrictions already proved in set theory; these are *not* new surreal theorems in substance. Benhamou, Cummings, Goldberg, Hayut, and Poveda prove that a singular strong-limit cardinal of uncountable cofinality cannot be the first such disagreement, and, from a measurable cardinal, obtain a model whose first disagreement is  $\aleph_\omega$  [16]. Hence their first theorem gives:

**Corollary 11.4** (Translation of a cited preprint theorem [04]). *If  $\lambda_H$  is singular and strong limit, then  $\text{cf}(\lambda_H) = \omega$ .*

*Proof.* An infinite singular cardinal has infinite cofinality. Uncountable cofinality is excluded by the cited compactness theorem, so its cofinality must be  $\omega$ .  $\square$

The preprint was submitted on August 25, 2026. It is important not to replace the cardinal conclusion of [theorem 11.2](#) by an unsupported regular-cardinal claim: the cited relative consistency result explicitly permits a singular first threshold. Source 04 credited only “the older repository discussion of the first new birthday across universes” in the birthday-cutoff report; the exact theorem is `univ:prop:beta`, and the same statement for transitive same-ordinal models is `hset:thm:firstbirthday`.

## 12 Complexity-bounded definability and escape ordinals

The unrestricted phrase “all parameter-free definable numbers” is external. A fixed syntactic complexity bound, by contrast, admits an internal treatment. This gives a precise version of the familiar intuition that definitions of bounded complexity can be diagonalized against without creating a paradox.

### 12.1 The bounded hierarchy [04]

Fix a standard natural number  $n \geq 1$ . Use a standard syntactic coding of formulas in the Lévy hierarchy, with padding so that  $\Sigma_n$  formulas are also admitted at every higher level. There is a partial truth formula  $\text{Sat}_{\Sigma_n}(e, x)$  for this fixed fragment. The existence of each fixed-fragment truth definition is standard; it does not provide one truth formula for all values of  $n$  simultaneously [10, 12].

**Definition 12.1** ([04]). Let  $S_n$  be the collection of unique solutions of parameter-free  $\Sigma_n$  formulas, and put  $B_n = S_n \cap \mathbf{No}$ . More explicitly,  $x \in S_n$  means that some code  $e$  of a formula with one free variable satisfies

$$\text{Sat}_{\Sigma_n}(e, x) \wedge \forall y (\text{Sat}_{\Sigma_n}(e, y) \longrightarrow y = x).$$

Each formula is required to have a unique solution among all sets, not just among the surreals.

**Lemma 12.2** ([04]). *For every fixed standard  $n$ , both  $S_n$  and  $B_n$  are parameter-free definable sets of ZFC.*

*Proof.* For a formula code having a unique solution, send the code to that solution; send every other code to a fixed default set. The fixed-fragment truth formula defines this total function on a subset of  $\omega$ , so Replacement gives its range as a set. Separation removes the default value when necessary and restricts to actual unique solutions. A further Separation restricts to surreal codes. All definitions use the fixed standard integer  $n$  only through its finite numeral, so there are no free parameters.  $\square$

**Theorem 12.3** (Definability-complexity escape [04]). *For every fixed standard  $n \geq 1$ , let*

$$\delta_n = \sup\{b(x) + 1 : x \in B_n\}, \quad d_n = \text{the all-plus surreal of length } \delta_n. \quad (12.1)$$

*Then  $d_n$  is a parameter-free definable omnific integer, but  $d_n \notin B_n$ . There is a standard  $m > n$  with  $d_n \in B_m$ . Consequently the hierarchy  $(B_n)$  does not eventually stabilize.*

*Proof.* Replacement gives the set of birthdays and the ordinal supremum in (12.1). The all-plus sequence is then uniquely defined, so  $d_n$  is parameter-free definable and is an ordinal surreal. If  $d_n \in B_n$ , the definition of the supremum would imply

$$b(d_n) + 1 = \delta_n + 1 \leq \delta_n,$$

which is impossible. On the other hand, the parameter-free formula just given for  $d_n$  is a particular finite formula. After conversion to prenex form and adding dummy quantifiers, it belongs to some  $\Sigma_m$  fragment. Thus  $d_n \in B_m$ . By nesting and its failure to lie in  $B_n$ , we must have  $m > n$ .  $\square$

*Remark 12.4* (No unproved complexity estimate [04]). The theorem does not assert  $m = n + 1$ , a closed expression for  $\delta_n$ , or a computable method that outputs the full transfinite sign sequence of  $d_n$ . The syntactic complexity of the normal-form, birthday, and sign-coding formulas depends on coding conventions. Only existence of a finite higher level is used. Nor is the sequence of escape ordinals asserted to be cofinal in all ordinals.

## 12.2 Why the separate definitions cannot be collected uniformly

**Theorem 12.5** (Noncollection [04, 07]; bounded clause for arbitrary  $A$  [merge]). *Let  $M$  be a model or the universe, and  $A$  an allowed parameter family.*

- (i) *There is no ambiently  $A$ -definable set  $S \subseteq \mathbf{No}$  such that  $K_A \subseteq S$ . In particular, no ambiently  $A$ -definable map  $\omega \rightarrow \mathbf{No}$  has range containing all of  $K_A$  [04, 07].*
- (ii) *No ambiently  $A$ -definable set contains every  $A$ -definable omnific integer in  $(0, \omega)$  [07].*

*Proof.* (i) If such an  $S$  were given, the simplest strict upper bound  $u = \{S \mid \emptyset\}$  is uniquely definable from  $S$ , so  $u \in K_A$ , but  $u > S$ , so  $u \notin S$  (07). Source 04 uses instead the all-plus surreal of the least ordinal strictly above all birthdays in  $S$ , whose birthday exceeds that of every member of  $S$ . For a definable map on  $\omega$ , Replacement gives an  $A$ -definable range set, reducing to the first assertion.

(ii) Restrict a putative set  $S$  to  $\text{ran}(\mathbf{Q})$  and apply  $\mathbf{Q}^{-1}$  pointwise. Separation and Replacement produce an  $A$ -definable set of surreals  $S' = \{\mathbf{Q}^{-1}(z) : z \in S \cap \text{ran}(\mathbf{Q})\}$ . Every  $x \in K_A$  has  $\mathbf{Q}(x) \in I_A \cap (0, \omega)$  by theorem 6.8, so the assumed property of  $S$  would give  $x \in S'$ . This contradicts (i).  $\square$

Source 07 stated the theorem for a finite parameter tuple  $p$  of a transitive  $M$ ; the proof uses only that the sets involved are defined from finitely many allowed parameters, so it holds for any  $A$ . Thus external countability does not even give an internally definable list of the definable omnific integers in this bounded interval. The theorem concerns *sets* of  $M$ , not a universal assertion that  $K_A$  cannot be a definable class: in a pointwise definable model it is simply  $\mathbf{No}^M$ , which is a definable class (07).

**Corollary 12.6** (Failure of a uniform bounded-truth collector [04]). *In an  $\omega$ -standard model of ZFC there is no single parameter-free formula that, for each  $n \geq 1$ , uniquely defines the set  $B_n$  above, uniformly in  $n$ .*

*Proof.* A uniform formula would, by Replacement, give a parameter-free set  $B = \bigcup_{n \geq 1} B_n$ . Every parameter-free definable surreal is uniquely defined by a finite formula and hence belongs to some  $B_n$ . Thus  $B$  would contain  $K_\emptyset$ , contrary to theorem 12.5. The  $\omega$ -standardness hypothesis ensures that internal natural indices are precisely the standard finite levels discussed here.  $\square$

This is the exact point at which the false “least undefinable surreal” argument tries to use more truth than it has. Separate formulas defining each fixed  $B_n$  do not constitute one internal formula defining the sequence of all these sets. The argument also does not claim that the membership class of definable elements can never accidentally be definable: in a pointwise definable model it is simply the entire universe. What fails there is an internal surjective set-sized list of those elements, not the external fact that every one of them has some definition.

### 13 Forcing can change the theory without adding reals

We now give a concrete source of models exhibiting the dichotomy. The forcing facts are standard, but proofs of the relevant homogeneity and closure steps are included. The result connects ambient definability of real numbers to genuinely transfinite surreal information without conflating the two.

**Lemma 13.1** (Homogeneity over a constructible ground [04]). *Let  $W$  be a transitive model of ZFC satisfying  $V = L$ , and let  $G$  be generic for an ordinal-definable weakly homogeneous forcing  $\mathbb{P} \in W$ . Then, in  $W[G]$ ,*

$$\text{HOD}^{W[G]} = W.$$

*The assertion has the usual relative-model interpretation; it does not assume that a generic filter over the actual universe is a set in that universe.*

*Proof.* First let  $Y \subseteq \theta$  be ordinal definable in  $W[G]$ . Membership in  $Y$  is expressed there by one formula  $\psi(\beta, \bar{\alpha})$  with a finite ordinal tuple. For each  $\beta < \theta$ , weak homogeneity implies that the top condition decides  $\psi(\check{\beta}, \check{\alpha})$ . Indeed, if one condition forced the sentence and another forced its negation, an automorphism could make their images compatible, while fixing all check names for ordinals. Compatible conditions cannot force contradictory sentences. Thus

$$Y = \{\beta < \theta : 1_{\mathbb{P}} \Vdash \psi(\check{\beta}, \check{\alpha})\} \in W$$

by the definability of the forcing relation.

Every HOD set is coded within HOD by a set of ordinals: use its canonical HOD well-order on a transitive closure and the pointed membership-relation code. Such a code is itself ordinal definable, so the previous paragraph places it in  $W$ . The absolute Mostowski collapse in  $W$  then recovers the original set. This proves  $\text{HOD}^{W[G]} \subseteq W$ . Conversely, forcing does not change the ordinals or the constructible hierarchy, so  $L^{W[G]} = W$ . Every constructible set is ordinal definable in  $W[G]$  using its position in the canonical constructible well-order, and so are all members of its transitive closure. Therefore  $W \subseteq \text{HOD}^{W[G]}$ . These are the standard homogeneity/HOD arguments [10].  $\square$

**Theorem 13.2** (Prescribed regular first birthday [04]). *Let  $W \models \text{ZFC} + V = L$  be transitive, and let  $\kappa$  be an infinite regular cardinal of  $W$ . Force with*

$$\text{Add}(\kappa, 1)^W = \{p : p \text{ is a partial map } \kappa \rightarrow 2, |\text{dom}(p)|^W < \kappa\},$$

*ordered by extension, and let  $V = W[G]$ . Then*

- (i)  $\text{HOD}^V = W$  and  $K_{\text{HOD}}^V = \mathbf{No}^W$  as ordered fields;
- (ii) the least birthday of a surreal outside  $K_{\text{HOD}}^V$  is exactly  $\lambda_H^V = \kappa$ ;
- (iii)  $\mathcal{U}^V$  contains the explicit ordinal-indexed independent family of [theorem 8.3](#) over the old field  $\mathbf{No}^W$ ;

- (iv) if  $\kappa > \omega$ , then  $\mathbb{R}^V = \mathbb{R}^W$ , and every real of  $V$  is ordinal definable in  $V$ , despite (ii) and (iii).

*Proof.* For two conditions  $p, q$ , flip the bits on the set of common-domain coordinates at which they disagree. This is a ground-model automorphism sending  $p$  to a condition compatible with  $q$ . Thus the forcing is weakly homogeneous and ordinal definable, so [lemma 13.1](#) applies. The field identification follows from [theorem 5.1](#) and the absoluteness of canonical surreal arithmetic.

Regularity of  $\kappa$  makes the union of a descending chain of length less than  $\kappa$  a condition. Given a name for a sequence of ordinals of length  $\mu < \kappa$ , recursively strengthen a condition to decide each coordinate, taking lower bounds at limit stages and at the end. The resulting ground-model sequence is forced to equal the name. Hence the forcing adds no subsets of any  $\mu < \kappa$ . For  $\kappa = \omega$ , the corresponding assertion about finite subsets is immediate. This closure also prevents a collapse of the cardinal  $\kappa$  by a shorter sequence.

The union  $g = \bigcup G$  is a total subset of  $\kappa$ , and is new: for every ground-model  $a \subseteq \kappa$ , the set of conditions already disagreeing with  $a$  at some coordinate is dense. Since  $\text{HOD}^V = W$ , the first non-HOD subset therefore occurs exactly at  $\kappa$ . [Theorem 11.2](#) proves (ii). The new sign sequence with positive positions  $g$  lies outside  $K_{\text{HOD}}^V$ ; normalize its absolute value as in [theorem 8.2](#) to obtain the parameter  $t$  for (iii).

When  $\kappa > \omega$ , the no-new-short-sequences conclusion in particular gives no new subsets of  $\omega$ , hence no new reals. All old reals belong to  $W = \text{HOD}^V$ , which proves (iv).  $\square$

**Example 13.3** (All reals ordinal definable, but many omnific integers not [04]). Take  $\kappa = \omega_1^W$  in [theorem 13.2](#). In the extension, every real is ordinal definable; every surreal of birthday less than  $\omega_1^W$  is in the HOD-surreal field; and a non-HOD surreal first appears at birthday  $\omega_1^W$ . Nevertheless a single fixed interval of purely infinite omnific integers already contains independent sets of every internal cardinality over the old surreal field. This is a strict separation between the real question and the surreal question, not a consequence of adding an undefinable real.

**Example 13.4** (A Cohen real [07]). Let a transitive ground universe satisfying  $V = L$  be extended by a Cohen real  $c$ . In the extension  $\text{HOD} = L$  (this is [lemma 13.1](#) for Cohen forcing, which is ordinal-definable and weakly homogeneous; the standard argument concerns OD subsets of ordinals and does *not* assert that homogeneous forcing adds no new OD sets of arbitrary kind). Consequently

$$z_c = \omega^{3/4} + \sum_{n < \omega} (1 + \chi_c(n)) \omega^{1/(n+2)}$$

is a non-OD omnific integer in  $(0, \omega)$ . The displayed support is parameter-free definable and every coefficient is either 1 or 2. This realizes [corollary 9.4](#) concretely, relative to the existence of the indicated ground model and generic extension. More generally, if an extension adds a non-OD subset  $X$  of an ordinal  $\alpha$ , [theorem 9.3](#) creates a non-OD omnific integer below  $\omega$ , even when that forcing adds no real numbers. This is an immediate conditional application of the code, not a claim that every forcing extension has the same HOD.

*Remark 13.5* (What is, and is not, absolute [04]). The forcing theorem compares old sign sequences and their arithmetic. It does not assert that every old parameter-free definition keeps its truth or uniqueness in an extension. An ambient definition can mention the continuum, a generic object, or the set-theoretic universe itself. Its truth can change even when the particular sign sequence and all arithmetic performed on old inputs stay unchanged. Nor is the relation  $V = \text{HOD}$  preserved by arbitrary forcing.

## 14 Relativization and a surcomplex companion

### 14.1 An allowed set of ordinal information [04]

Let  $p$  be a fixed set of ordinals. Write  $\text{OD}(p)$  for definitions using  $p$  together with finitely many ordinals, and  $\text{HOD}(p)$  for hereditary membership in this class. Here the parameter is the *single set*  $p$ , not permission to use its arbitrary elements as unrelated parameters. Since  $p$  is a set of ordinals, it belongs to  $\text{HOD}(p)$ ; this is an inner model of ZFC with its corresponding canonical definable well-order [10].

**Proposition 14.1** (Relativized theory [04]). *Put  $K_p = \mathbf{No} \cap \text{OD}(p)$  and  $I_p = \mathbf{Oz} \cap \text{OD}(p)$ . Then*

$$K_p = \mathbf{No} \cap \text{HOD}(p) = \mathbf{No}^{\text{HOD}(p)}, \quad K_p = \text{Frac}(I_p).$$

*The field  $K_p$  is real closed and  $I_p$  is an integer part. Every member of  $\mathcal{U}$  is in  $\text{OD}(p)$  if and only if  $V = \text{HOD}(p)$ . If  $V \neq \text{HOD}(p)$ , the family*

$$\omega + \omega^{t\omega^{-(\alpha+1)}} \quad (\alpha \in \text{Ord})$$

*is algebraically independent over  $K_p$  for every  $t \in (0, 1) \setminus K_p$ .*

*Proof.* The hereditary sign-code argument only uses that ordinals and the fixed sign tags are hereditarily definable; permitting  $p$  does not change it. Unique-output closure gives the field, floor, and fraction claims. The reversible code uses no extra parameter, while the membership-graph collapse works inside  $\text{HOD}(p)$  because that inner model contains  $p$ . Finally every normal-form entry of an  $\text{OD}(p)$  surreal is definable from  $p$  and ordinal parameters, so its support lies in  $K_p$ . The support-subfield amplification proof applies without change.  $\square$

This relativization describes precisely how adding allowed information changes the theory. It is different from allowing *all* surreal parameters, which makes every individual surreal trivially definable by equality to itself.

### 14.2 Canonical surcomplex pairs [04, 07]

For completeness, define the surcomplex field by pairs  $\mathbf{No}(i) = \mathbf{No} \times \mathbf{No}$  with  $i = (0, 1)$  and the usual quadratic extension operations. This fixed coordinate representation names a real axis and an imaginary unit.

**Proposition 14.2** (Definable surcomplex closure [04, 07]). *In the canonical pair representation, the ambiently  $A$ -definable surcomplex numbers are exactly  $K_A(i)$ , and they form an algebraically closed field. The ordinal-definable ones are exactly  $K_{\text{HOD}}(i)$ . If  $V \neq \text{HOD}$ , the real omnific family in [theorem 8.3](#) remains algebraically independent over  $K_{\text{HOD}}(i)$  inside  $\mathbf{No}(i)$  [04].*

*Proof.* Pairing and the two coordinate projections are parameter-free set-theoretic operations. Thus a pair is  $A$ -definable precisely when both coordinates are. The quadratic extension of a real closed field by  $i$  is algebraically closed. For the independence assertion, write a purported complex-coefficient polynomial relation as  $P_0(\bar{z}) + iP_1(\bar{z}) = 0$ , with polynomials over  $K_{\text{HOD}}$ . At real inputs both values are real, so both are zero; a nonzero relation would contradict real algebraic independence.  $\square$

There is no conflict with complex conjugation or with the failure of a pure field to choose an imaginary unit. The proposition uses *ambient canonical coordinates* with their distinguished orientation. An abstract algebraically closed-field language without those coordinates has a different definable closure, just as the abstract real field does; coordinate-based definitions are not silently transferred to that reduct.

## 15 Changing the language changes the answer

The pure ordered field is [theorem 3.4](#).

### 15.1 Adding the ordinary exponential still does not define omega [07]

**Proposition 15.1** ([07]). *In the exponential-field reduct,*

$$\text{dcl}_{\mathbf{No}_{\text{exp}}^M}(\emptyset) \subseteq \mathbb{R}^M.$$

*In particular  $\omega$  is not parameter-free definable in that reduct.*

*Proof.* The embedded real exponential field is elementary in the surreal exponential field by the classical theorem recalled in [section 2.5](#). A parameter-free formula having a unique solution in the larger structure must have a solution in the elementary substructure, and uniqueness makes these the same solution. Thus the solution is real.  $\square$

We make no attempt to identify the exact parameter-free definable closure in the real exponential field. In particular, this argument supplies no conclusion about whether any particular familiar transcendental real beyond those directly named by the language is definable there.

### 15.2 Simplicity already defines omega [07]

**Proposition 15.2** ([07]). *In the structure  $(\mathbf{No}^M; <, \prec_s)$ , the ordinal subclass and the element  $\omega$  are parameter-free definable.*

*Proof.* The condition

$$\text{OrdSign}(x) : \quad \forall y (y \prec_s x \longrightarrow y < x)$$

holds exactly for all-plus sequences. It holds for an ordinal because each proper prefix is smaller. If  $x$  has a minus sign, the prefix ending just before that sign is greater than  $x$ , so the condition fails. On this definable ordinal subclass, the numerical order is the usual ordinal order. A nonzero ordinal is a limit exactly when it has no greatest ordinal predecessor; this is first-order expressible by

$$x > 0 \quad \wedge \quad \forall y (\text{OrdSign}(y) \wedge y < x \longrightarrow \exists z (\text{OrdSign}(z) \wedge y < z < x)).$$

The least such ordinal is  $\omega$ .  $\square$

The formula  $\text{OrdSign}$  is not new: the birthday-cutoff report uses the same formula, “every proper prefix of  $x$  lies below  $x$ ”, in the proof of `hset:prop:nondet`, which shows that neither birthday nor prefix simplicity is definable in the ordered-field reduct or any o-minimal expansion. Source 07 did not credit it. Much stronger interpretations become possible after adding birthday structure: the birthday-cutoff report develops birthday-enriched interpretations and bi-interpretations [20], and Chen–Hamkins–Yang’s related work has been publicly announced by Hamkins [17]. None of those interpretation results is needed to prove the coding and core theorems above, and none is asserted here as a new result.

### 15.3 The field with the omnific predicate: what the collection proves [merge]

Both sources asked what becomes definable in  $\mathfrak{S} = (\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$ , the field with only the omnific predicate added (04's Question 1, 07's question on intermediate languages). At their pin the collection did not address this. It now contains results that largely answer it; we record them with proofs from the cited theorems.

**Proposition 15.3** (The pair  $(\mathbf{No}, \mathbf{Oz})$  [merge]).

- (i)  $\text{dcl}_{\mathfrak{S}}(\emptyset) \subseteq \mathbb{R}$ . In particular no infinite or infinitesimal surreal, and so neither  $\omega$  nor any code value of [section 6](#), is parameter-free definable in  $\mathfrak{S}$ .
- (ii)  $\mathbb{R}$  and  $\mathbb{Z}$  are parameter-free definable subsets of  $\mathfrak{S}$ . Hence every real that is parameter-free definable in  $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$  belongs to  $\text{dcl}_{\mathfrak{S}}(\emptyset)$ ; since that structure interprets second-order arithmetic, these include every real whose binary expansion is parameter-free definable in second-order arithmetic.
- (iii) For every set  $P \subseteq \mathbf{No}$ , neither the class of Conway monomials nor the graph of  $x \mapsto \omega^x$  is definable in  $\mathfrak{S}$  with parameters from  $P$ .

*Proof.* (i) Every field automorphism of  $\mathbf{No}$  preserves the order, since the positive elements are the nonzero squares; so a field automorphism preserving  $\mathbf{Oz}$  is an automorphism of  $\mathfrak{S}$ . Fix a formula  $\varphi(v)$  with a unique solution  $x$  in  $\mathfrak{S}$ . By induction on the subformulas of this fixed formula, every class automorphism  $\sigma$  of  $\mathfrak{S}$  preserves  $\varphi$ , so  $\sigma(x) = x$ . By `opa:thm:fixed` of the automorphisms report [24], the common fixed field of all field automorphisms of  $\mathbf{No}$  preserving  $\mathbf{Oz}$  is  $\mathbb{R}$ . Hence  $x \in \mathbb{R}$ . (The dilation  $\mathbf{S}_2$  of `odg:def:ex:dilation`,  $\sum_g a_g \omega^g \mapsto \sum_g a_g \omega^{2g}$ , already suffices: it is a field automorphism preserving  $\mathbf{Oz}$ , and it fixes  $x$  only if  $2 \text{supp } x = \text{supp } x$ ; for a reverse well-ordered set this forces  $\text{supp } x \subseteq \{0\}$ , since a nonzero member  $g$  would give the increasing sequence  $g, 2g, 4g, \dots$  or  $g, g/2, g/4, \dots$  inside the support.)

(ii) By `odg:def:thm:realrecovery` and `odg:def:cor:internal` of the omnific Diophantine report [22], the pure ring  $\mathbf{Oz}$  parameter-free interprets  $\mathbf{No}$  as its fraction field and defines in it the subfield  $\mathbb{R}$  by a formula whose quantifiers range over  $\mathbf{Oz}$ . In  $\mathfrak{S}$  the map  $(a, b) \mapsto a/b$  on  $\mathbf{Oz} \times (\mathbf{Oz} \setminus \{0\})$  is definable, so relativizing that formula to  $\mathbf{Oz}$  defines  $\mathbb{R} \subseteq \mathbf{No}$  without parameters. Then  $\mathbb{Z} = \mathbb{R} \cap \mathbf{Oz}$ . A parameter-free definition in  $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$  relativizes to one in  $\mathfrak{S}$ . That structure defines  $\mathbb{N}$  and codes subsets of  $\mathbb{N}$  by binary expansions of reals, which gives the last clause. (The pure ring  $\mathbf{Oz}$  already defines  $\mathbb{Z}$  and  $\mathbb{N}$ , `odg:def:cor:arithmetic`.)

(iii) This is `opa:thm:parameters` of the automorphisms report.  $\square$

So among ambiently definable surreals only reals can become parameter-free definable after adding the omnific predicate, and the codes of [section 6](#), which recognize ordinal-indexed monomials and normal-form coefficients through set theory, have no counterpart there. The exact identification of  $\text{dcl}_{\mathfrak{S}}(\emptyset)$  inside  $\mathbb{R}$  is not known; it is squeezed between the reals definable in  $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$  and  $\mathbb{R}$ . The expansion by the omega map is not addressed by these results. [Question 17.1](#) is re-scoped accordingly.

## 16 Arithmetic of the definable omnific ring

These results of source 07 apply to  $I = I_A$  and, with the same proofs, to the OD omnific class. Their full-class antecedents are in the omnific Diophantine and quotient reports [22, 23], named at each statement below; the arguments are included to verify that the required division and support operations preserve ambient definability. They are restricted duplicates, not new results.

**Proposition 16.1** (The constant-term retraction [07]). *There is a split exact sequence of additive groups and a ring retraction*

$$0 \longrightarrow \Pi_A \longrightarrow I_A \xrightarrow{\text{ct}} \mathbb{Z} \longrightarrow 0, \quad I_A = \Pi_A \oplus \mathbb{Z}.$$

The ideal  $\Pi_A$  is divisible as an additive group. For every positive ordinary integer  $n$ ,

$$nI_A = \Pi_A \oplus n\mathbb{Z}, \quad I_A/nI_A \simeq \mathbb{Z}/n\mathbb{Z}. \quad (16.1)$$

Furthermore  $\Pi_A^2 = \Pi_A$ .

*Proof.* The purely infinite part and the constant coefficient are uniquely definable from the normal form, so both remain in  $K_A$ . Products of terms with positive exponents still have positive exponents; therefore constant term is a ring homomorphism and  $\Pi_A$  is an ideal. For  $p \in \Pi_A$ , division by  $n$  changes coefficients but not positive support, so  $p/n \in \Pi_A$ . This proves divisibility and (16.1).

For the final assertion take nonzero  $p \in \Pi_A$  with exponent set  $S \subseteq \mathbf{No}_{>0}$ . The simplest separator  $\delta = \{0 \mid S\}$  is positive and strictly below every exponent of  $p$ . It is definable from  $p$ . Factor  $p = \omega^\delta(p\omega^{-\delta})$ . Both factors have strictly positive normal-form exponents and belong to  $K_A$ , so both are in  $\Pi_A$ . Thus  $\Pi_A \subseteq \Pi_A^2$ ; the reverse containment follows because it is an ideal. The zero case is automatic.  $\square$

The full-class statement is `odg:thm:finitequotients` ( $n\mathbf{Oz} = \Pi + n\mathbb{Z}$ ,  $\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}$ ).

**Theorem 16.2** (All finite quotients [07]). *Every unital ring homomorphism from  $I_A$  to a finite ring factors through  $\text{ct} : I_A \rightarrow \mathbb{Z}$ . Every finite-index ideal of  $I_A$  is  $nI_A$  for a unique positive integer  $n$ . Consequently the profinite completion of  $I_A$  is canonically  $\widehat{I_A} \simeq \widehat{\mathbb{Z}}$ . Also  $\text{ct}$  is the unique unital ring homomorphism  $I_A \rightarrow \mathbb{Z}$ .*

*Proof.* Let  $f : I_A \rightarrow R$  and suppose the additive group of  $R$  has finite exponent  $m$ . For  $p \in \Pi_A$  choose  $q \in \Pi_A$  with  $p = mq$ . Then  $f(p) = mf(q) = 0$ . Thus  $f$  factors through  $I_A/\Pi_A \simeq \mathbb{Z}$ . If an ideal  $J$  has finite index, apply this argument to  $I_A \rightarrow I_A/J$  to get  $\Pi_A \subseteq J$ . The image of  $J$  in  $\mathbb{Z}$  is  $n\mathbb{Z}$  for a unique  $n \geq 1$ . Hence  $J = \Pi_A \oplus n\mathbb{Z} = nI_A$ . The compatible quotient identifications in (16.1) identify the inverse system of all finite quotients with that of  $\mathbb{Z}$ . For a homomorphism  $f : I_A \rightarrow \mathbb{Z}$ , the image of each  $p \in \Pi_A$  is divisible by every positive integer because  $p$  is. The only such integer is zero. Again  $f$  factors through  $\text{ct}$ , and unitality forces its restriction to  $\mathbb{Z}$  to be the identity.  $\square$

For the full ring these are `odg:thm:finitequotients`, `odg:cor:charideals` and `odg:eq:profinite` of the omnific Diophantine report, and the much stronger universal theorem `osq:thm:universal` of the quotient report, which factors every homomorphism into a set-sized ring through  $\text{ct}$ . Related (batch 34): the surcomplex-automorphism report proves the finite-quotient statement, and the transfer of solvability of integer polynomial systems through  $\text{ct}$ , uniformly for its integer-part rings  $R_F$  over set-sized real closed fields  $F$ , with  $\mathbf{Oz} = R_{\mathbb{R}}$  and  $2^c$  pairwise nonisomorphic rings among them (`saut:gr:prop:finite`, `saut:gr:prop:dio`); it cites the theorem above for the rings  $I_A$ .

**Proposition 16.3** (A size-sensitive nontransfer [07]; already in the quotient report). *For a transitive set model  $M$  and finite  $A$ , the externally countable ring  $I_A^M$  admits a homomorphism to a set-sized ring which does not factor through constant term.*

*Proof.* Take the identity map  $I_A^M \rightarrow I_A^M$ . Its target is a set externally, and is countable by (3.1). It cannot factor through constant term because  $\omega \in I_A^M$ ,  $\omega \neq 0$ , and  $\text{ct}(\omega) = 0$ .  $\square$

There is no contradiction with the full-class theorem. Its source is the entire proper class of omnific integers in the same universe in which target size is measured. Here the source is a fragment, and the external set  $I_A^M$  need not be an internal set of  $M$ . The finite-target theorem survives because its proof uses divisibility rather than a proper-class size obstruction. Source 07 presented this “external size-boundary warning” as supplied by it; it is already in the quotient report, in the paragraph “The universe-relative reading” of `osq:sub:foundations` (from that report’s manuscripts 04, 10 and 11), which observes that a set-sized fragment viewed in a larger universe has its own identity map as a set-sized target.

**Proposition 16.4** (Non-Noetherian behaviour [07]). *The ring  $I_A$  is not Noetherian. In fact it has the strict ascending chain*

$$(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/3}) \subsetneq \dots$$

*Proof.* Every displayed monomial is parameter-free ambient-definable. Since  $\omega^{1/n} = \omega^{1/(n+1)}\omega^{1/n-1/(n+1)}$  and the second factor is omnific, the ideals are increasing. Equality would put  $\omega^{1/(n+1)}/\omega^{1/n} = \omega^{-1/(n(n+1))}$  in  $I_A$ . This is a nonzero infinitesimal, hence is not omnific. Therefore each inclusion is strict.  $\square$

The full-ring chain  $(\omega) \subsetneq (\omega^{1/2}) \subsetneq (\omega^{1/4}) \subsetneq \dots$  is `odg:prop:notnormal`.

**Proposition 16.5** (Elementary Diophantine transfer [07]). *A finite system of polynomial equations with ordinary integer coefficients has a solution in  $I_A$  if and only if it has a solution in  $\mathbb{Z}$ . This equivalence need not persist when nonvanishing conditions are added.*

*Proof.* Apply the ring retraction `ct` coordinatewise to any solution in  $I_A$ . Conversely every integer solution already belongs to  $I_A$ . For the failure with nonvanishing, the equation  $X^2 = 2Y^2$  has the nonzero omnific solution  $X = \sqrt{2}\omega$ ,  $Y = \omega$ , both parameter-free ambient-definable, whereas the only ordinary integer solution is  $(0, 0)$ .  $\square$

The full-ring statements are `odg:thm:transfer` and `odg:def:thm:ideal`, where  $\exists y x^2 = 2y^2$  defines  $\Pi$  in **Oz**. In particular the nonnegative part of  $I_A$  is not a model of Peano arithmetic: PA proves that the displayed equation has no nonzero solution. This is distinct from open induction ([corollary 4.6](#)), the familiar weaker theory associated with integer parts of real closed fields [11]; no stronger induction assertion is being inferred from finite-quotient agreement with  $\mathbb{Z}$ .

*Remark 16.6* (Relation to `odg:q:size` [merge]). The omnific Diophantine report asks (`odg:q:size`) which set-sized exponent groups or universe-relative constructions admit analogues of the common-monomial and set-sized quotient theorems. The definable fragments  $K_A$  and  $I_A$  supply one instance: same-parameter per-element monomial clearing holds ([theorems 4.1](#) and [10.5](#)); finite quotients remain  $\mathbb{Z}/n\mathbb{Z}$  ([theorem 16.2](#)); and the set-sized quotient theorem fails through the identity map ([proposition 16.3](#), already observed in the quotient report). This is partial information only. The question is about general set-sized exponent groups and constructions, and it stays open.

## 17 Foundational boundaries, further questions, and formalization

### 17.1 From set models to an intended universe [07]

The set-model convention is a clean semantic setting, not a theorem of ZFC that such a model exists. All theorems quantified over such  $M$  are conditional statements. The set-theoretic coding

constructions themselves, including  $\mathbf{C}$ ,  $\mathbf{T}$ ,  $\mathbf{Q}$ ,  $\mathbf{Q}_{a,b}$ ,  $E_\alpha(X)$  and  $\text{esc}$ , are ordinary ZFC-definable class constructions and do not require the existence of a set model.

For an intended universe  $V$ , one may discuss parameter-free definability externally, or work in a stronger metatheory supplying the relevant truth apparatus. What one may not do is count all true definitions of  $V$  using an unmentioned truth predicate and then invoke Replacement *inside that same*  $V$ . The OD/HOD part avoids this difficulty because OD and HOD have internal first-order definitions. The noncollection theorem is formulated precisely so that no hidden internal list of definitions is used.

For nonstandard models of ZFC, [definition 3.1](#) still uses standard external formulas, and the closure-under-definitions principle continues to hold; so do the pointwise test and the coding theorems, whose proofs are internal. However, the ordinal and sign arguments about a *least externally omitted ordinal* need not transfer unchanged: the model's ordinals may not be externally well founded. [Theorem 10.3](#) and its corollaries are not claimed for such models. The pointwise-definability literature treats nonstandard models as well as transitive cases [\[12\]](#); these distinctions should be retained when importing its results.

## 17.2 Questions

The questions below are further projects, not omitted hypotheses or unfinished steps in the proofs above. They are not asserted to be previously published open problems.

**Question 17.1** (Languages between fields and set theory [\[04, 07\]](#); re-scoped). Determine the parameter-free definable closure of  $\mathfrak{S} = (\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$  inside  $\mathbb{R}$ , or of well-specified set-sized models of its first-order theory. How much of ambient definability is captured by an expansion naming the omega map, which bounded coding maps become definable there, and what are its intrinsic definable closures?

Source 04 asked the first part in the form “which ambiently definable surreals become intrinsically definable after adding only the omnific predicate”; source 07 asked about  $(\mathbf{No}; <, +, \cdot, \mathbf{Oz})$  and the omega-map expansion. By [proposition 15.3](#) (from the automorphisms and omnific Diophantine reports) only reals can, all reals definable in  $(\mathbb{R}; +, \cdot, <, \mathbb{Z})$  do, and the monomials are not definable with set parameters; the exact real closure and the omega-map expansion remain open. The pure ordered field defines only real algebraic elements without parameters ([theorem 3.4](#)); order plus simplicity already defines  $\omega$  ([proposition 15.2](#)); birthday-enriched interpretation results supply a much stronger endpoint, but do not settle every intermediate language by themselves.

**Question 17.2** (Birthday cost of reversible omnific coding [\[04\]](#)). Obtain sharp upper and lower bounds for  $b(\mathbf{C}(s))$  in terms of  $b(s)$ , and compare alternative reversible codes with the same leading term and coefficient alphabet. Can a useful optimality statement be proved for a specified class of such codes?

The construction controls support order type and the leading scale, not the exact surreal birthday of its nested monomials. No unstated birthday estimate is used in the independence theorem. [Lemma 2.6](#) bounds  $b(\omega^a)$  by  $\omega^{b(a)}$ , which is a starting point for upper bounds.

**Question 17.3** (Intermediate support fields [\[04\]](#)). Classify naturally occurring proper subfields satisfying [\(8.3\)](#), especially those arising from relative definability or canonical inner models. Determine when stronger structural invariants than the independent family can be read from the quotient additive group  $\mathbf{No}/K$ .

The support-subfield theorem already gives one invariant uniformly: a single element outside such a field produces an ordinal-indexed independent family in one fixed omnific interval.

Instances are  $K_{\text{HOD}}$  and  $K_p = \mathbf{No} \cap \text{OD}(p)$  (proposition 14.1). For a proper same-ordinal inner model  $M$  as in the universes report,  $\mathbf{No}^M$  would satisfy (8.3) if the omega map and normal forms were absolute between  $M$  and  $V$ ; `univ:prop:absolute` proves absoluteness of the arithmetic and of cuts only, so this instance is not claimed from that report. For inner models with the same reals it is now supplied by the large-cardinal report (remark 17.6).

**Question 17.4** (Possible first omitted ordinals [07]). For which transitive  $M$  and parameter sets  $A$  can a specified epsilon number occur as  $\rho_A^M$ ? What additional closure constraints follow from definable ordinal functions beyond finite Cantor normal form?

There are immediate further constraints:  $O_A^M$  is closed under every parameter-free definable ordinal-valued function on definable inputs. For example, if an ordinal index is definable, its indexed epsilon number is definable. Determining the exact resulting external cut is a set-theoretic problem, not a matter of adding a few field operations.

**Question 17.5** (Effective syntactic costs [07]). After fixing a proof system and a precise coding of formulas, how efficiently can one translate definitions through  $\mathbf{Q}$ ,  $E_\alpha$ , and the denominator rule? Which translations preserve a chosen level of the set-theoretic formula hierarchy?

Interdefinability is proved here by fixed formula substitution. It does not automatically preserve quantifier complexity, and no optimal description-length theorem is asserted. This question links the present semantic theory to the collection's separate work on computable surreals without identifying the two notions.

*Remark 17.6* (The same-reals instance, from the large-cardinal report). The report `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms` proves, in its absoluteness lemma (`lce:lem:absolute`, from three manuscripts of batch 30), that for transitive inner classes  $M \subseteq V$  of ZFC with the same ordinals and the same reals the inclusion  $\mathbf{No}^M \subseteq \mathbf{No}$  preserves the field operations, the omega map and canonical normal forms. Its `lce:rem:dsnsupport` draws the consequence for question 17.3: if moreover  $M \neq V$ , then  $K = \mathbf{No}^M$  satisfies (8.3) and is proper (a transitive inner model of ZFC containing every set of ordinals is  $V$ ), so theorem 8.2 applies to it. The targets of elementary embeddings  $j : V \rightarrow M$  with a critical point, such as the ultrapower by a normal measure, are instances. Inner models that lack some real, such as  $L$  when  $\mathbb{R} \not\subseteq L$ , or HOD when it lacks a real, are not covered, and the classification asked in question 17.3 stays open. This remark, and the sentence after question 17.3 that points here, were added when the large-cardinal report was written; nothing else in this report changed.

### 17.3 A realistic Lean decomposition

A formalization should separate the general Hahn-field mathematics from satisfaction and internal models (04). The shortest algebraic route is to formalize lemma 8.1 for set-sized Hahn series and finite polynomials first: coefficient supports in different cosets cannot cancel. Next formalize the finite linear independence of monomials, the normalized-parameter argument, and the finite form of theorem 8.2. These steps require no model-theoretic truth predicates or proper-class objects.

The set-theoretic track then needs a precise universe convention, canonical sign codes, unique normal forms, and definable class maps. HOD identification requires a model-of-ZFC interface, with standard external formulas kept distinct from internal formula codes. The statements about  $B_n$  should be separate theorems for each fixed fragment, not a falsely uniform truth evaluator. Source 07 proposes three interfaces: a sign/normal-form interface with definable canonical operations; an external definability-closure interface for set structures; and a model-theoretic selection interface for o-minimal exponential fields. The bounded code uses only the first

two, while the core theorem additionally uses the published epsilon-cutoff theorem and the intersection argument. The repository documents sign, Hahn, size, and normal-form interfaces, but neither source nor this report maps the new statements to checked declarations, claims that the required interfaces already exist in Lean, or claims to have built or extended the Lean code [19].

## 18 Conclusion

There is no single language-independent class of “the definable surreals.” Once the ambient universe and permitted parameters are fixed, however, the basic algebra is robust: definable surreals form a real closed field, exponential and elementary over a transitive model, and definable omnific integers are an integer part with exactly that fraction field. Ordinal parameters reveal a particularly natural inner field, the canonical surreal field of HOD.

The strongest results are not counting observations. Explicit normal-form maps compress every surreal definability degree into a fixed-leading-term omnific interval, or into single monomials below  $\omega$ . If HOD fails to contain the universe, one missing surreal generates an ordinal-indexed algebraically independent family in that same interval. The failure of the definable field to be initial is controlled exactly by the first omitted ordinal, and the resulting maximal initial fragment is still an elementary exponential field and the fraction field of its omnific integers. Uniform coefficient assignments, rather than small coefficient alphabets, numerical size or ordinary congruences, carry the missing information. These conclusions coexist with, rather than evade, the traditional cautions about truth, parameters, HOD, and pointwise definable models.

## A Provenance, proof scope, and reproducibility

### A.1 Where each section comes from

Table 1: Provenance of the sections.

| Section                  | Sources                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Introduction           | 04’s introduction and 07’s scope section, merged; provenance and conventions added.                                                                              |
| 2 Foundations            | 04 §2 with 07 §2 (simplicity, initial classes, two exponentials, floor prefix, imported cutoff theorem and birthday controls); <a href="#">remark 2.5</a> merge. |
| 3 Definability           | 04 §3 with 07 §3; field dcl printed once.                                                                                                                        |
| 4 Field and integer part | 04 §4 with 07 §4 (selection, $\mathcal{L}_{\text{exp}}$ elementarity, cut denominator, valuation, normal-form invariant).                                        |
| 5 HOD                    | 04 §5 with 07 §8.1; proof of <a href="#">proposition 5.4</a> corrected.                                                                                          |
| 6 Codes                  | 04 §6 and 07 §§5.1–5.3.                                                                                                                                          |
| 7 Universe tests         | 04 §7 with 07 §§5.4, 8.2, 8.3.                                                                                                                                   |
| 8 Transcendence          | 04 §8.                                                                                                                                                           |
| 9 Uniformity             | 04 §9 with 07 §6.                                                                                                                                                |
| 10 Core                  | 07 §7.                                                                                                                                                           |
| 11 First birthday        | 04 §10, printed by citation of the universes report.                                                                                                             |
| 12 Complexity            | 04 §11 with 07 §5.5.                                                                                                                                             |
| 13 Forcing               | 04 §12 with 07 §8.4.                                                                                                                                             |
| 14 Relativization        | 04 §13 with 07 §9.4.                                                                                                                                             |

| Section                       | Sources                                               |
|-------------------------------|-------------------------------------------------------|
| <a href="#">15</a> Languages  | 07 §§9.2–9.3; <a href="#">proposition 15.3</a> merge. |
| <a href="#">16</a> Arithmetic | 07 §10 with credits added.                            |
| <a href="#">17</a> Questions  | 04 §14 and 07 §11.                                    |
| Appendices                    | 04’s appendices and 07’s audit section and gallery.   |

## A.2 Claim ledger

Table 2: Status of the statement families.

| Statement family                                                    | Source | Status                      | Dependency and qualification                                                                                                                                                                                        |
|---------------------------------------------------------------------|--------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signs, normal forms, real closedness, omega map, exponential, floor | 04, 07 | Classical                   | Conway, Gonshor, van den Dries–Ehrlich (with erratum), L’Innocente–Mantova; elementary proofs supplied where useful.                                                                                                |
| Exponential elementarity and epsilon cutoffs                        | 07     | Published input             | van den Dries–Ehrlich, erratum, Bournez–Guilmant. Not new.                                                                                                                                                          |
| Ambient field and integer-part closure                              | 04, 07 | Standard-method consequence | Unique-output closure, ordered finite root selection, and a uniformly chosen denominator. No priority claim for the closure principle; $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$ is <code>odg:thm:fractions</code> . |
| $K_A^M \prec \mathbf{No}^M$ in $\mathcal{L}_{\text{exp}}$           | 07     | Application                 | Fixed-formula translation and o-minimal definable selection; transitive $M$ .                                                                                                                                       |
| Global birthday/set-theory interpretation                           | 04, 07 | Prior announced work        | Chen–Hamkins–Yang; not used as a black-box dependency and not claimed.                                                                                                                                              |
| HOD/sign-code identification                                        | 04, 07 | Foundational application    | Hereditariness of a sign code, inner-model recursion, canonical normal forms. No claim to have introduced HOD-valued surreal fields.                                                                                |
| Fixed-interval sign code                                            | 04     | Proposed contribution       | <a href="#">Theorem 6.2</a> ; search did not locate this precise code.                                                                                                                                              |
| Bounded and local monomial codes                                    | 07     | Proposed contribution       | <a href="#">Theorems 6.8</a> and <a href="#">6.9</a> , with two-way parameter preservation.                                                                                                                         |
| One-interval universe tests                                         | 04, 07 | Proposed consequences       | <a href="#">Theorems 7.2</a> and <a href="#">7.7</a> ; standard pointed set coding followed by the omnific codes. Density form is Hamkins 2012; pointwise model existence is HLR.                                   |

| Statement family                                                          | Source  | Status                          | Dependency and qualification                                                                                                                                                                     |
|---------------------------------------------------------------------------|---------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Support-subfield amplification and HOD dichotomy                          | 04      | Principal proposed contribution | <a href="#">Theorems 8.2</a> and <a href="#">8.3</a> ; finite support-coset proof. No transcendence-basis claim.                                                                                 |
| Two uniformity gaps                                                       | 04, 07  | Proposed explicit formulations  | <a href="#">Theorems 9.1</a> , <a href="#">9.3</a> and <a href="#">9.6</a> and <a href="#">proposition 9.5</a> ; the general distinction between pointwise and uniform definability is standard. |
| Maximal initial elementary core                                           | 07      | Principal proposed contribution | <a href="#">Theorems 10.3</a> and <a href="#">10.5</a> and <a href="#">corollary 10.6</a> ; birthday estimates and cutoffs are classical inputs; transitive $M$ .                                |
| Bounded birthday escape and noncollection                                 | 07 (04) | Proposed consequences           | <a href="#">Theorems 6.10</a> and <a href="#">12.5</a> .                                                                                                                                         |
| First non-HOD birthday                                                    | 04      | Duplicate, printed by citation  | <code>univ:prop:beta</code> with $M = \text{HOD}$ ; also <code>hset:thm:firstbirthday</code> ; the compactness restriction is [16].                                                              |
| Bounded-complexity escape                                                 | 04      | Standard-method refinement      | Fixed-fragment truth and Replacement; no all-level truth predicate and no asserted one-step complexity bound.                                                                                    |
| Forcing without new reals                                                 | 04, 07  | Application                     | Standard homogeneous closed forcing over $L$ , combined with the independent-family theorem.                                                                                                     |
| Density iff $V = \text{HOD}$                                              | 07      | Hamkins 2012                    | Credited in <a href="#">theorem 7.5</a> .                                                                                                                                                        |
| Old-field closed nowhere denseness                                        | 07      | Specialization                  | <code>univ:thm:nowheredense</code> .                                                                                                                                                             |
| Simplicity defines $\omega$                                               | 07      | Elementary                      | The formula is that of <code>hset:prop:nondef</code> .                                                                                                                                           |
| Pair ( <b>No</b> , <b>Oz</b> )                                            | merge   | From the collection             | <code>opa:thm:fixed</code> ,<br><code>opa:thm:parameters</code> ,<br><code>odg:def:thm:realrecovery</code> .                                                                                     |
| Finite quotients, constant term, non-Noetherian, equations, size boundary | 07      | Restricted duplicates           | <code>odg:thm:finitequotients</code> ,<br><code>odg:prop:notnormal</code> ,<br><code>odg:thm:transfer</code> ,<br><code>osq:thm:universal</code> ,<br><code>osq:sub:foundations</code> .         |
| Finite verification                                                       | 04, 07  | Regression tests only           | Finite symbolic codes, decoders, formal support-coset and floor checks. They prove no infinite or set-theoretic theorem.                                                                         |

### A.3 Stale statements and corrections

1. *“All 51 reports.”* 04’s inspection note said that it was not a review of “all 51 reports”. At the pin the collection had 51 reports; at the placement commit it has 53 written reports, and this report was placed together with two others.
2. *First new birthday.* 04 credited only an “older repository discussion” in the birthday-cutoff report. The exact theorem, with the same proof, is `univ:prop:beta`; the birthday-cutoff report has it for same-ordinal models as `hset:thm:firstbirthday`. 04’s theorem is printed by citation.
3. *Hamkins 2012.* 04 did not cite Hamkins’s density/ $V = \text{HOD}$  answer, which is close to its equivalence of (i) and (ii) in [theorem 7.2](#); the credit is added.
4. *Size boundary.* 07 described its size-boundary warning as “supplied here”. It is in the quotient report (`osq:sub:foundations`).
5. *OrdSign.* 07’s formula defining the ordinals from order and prefix is the one in the proof of `hset:prop:nondef`; credit added.
6. *Cutoff theorem.* 07 cited van den Dries–Ehrlich without the erratum; the erratum and Bournez–Guilmant are added.
7. *Universes report.* 07’s “Theorem 14.3 in the report” is `univ:thm:nowheredense`. 07’s statement that the universes report “develops this compatibility explicitly” (for normal forms and the exponential) is only partly accurate: `univ:prop:absolute` covers the arithmetic and cuts.
8. *Restricted duplicates.* 07’s arithmetic results are restricted versions of `odg:thm:finite` quotients, `odg:prop:notnormal`, `odg:thm:transfer`, `odg:def:thm:ideal` and the open-induction discussion `odg:rem:foursquares`; the labels are now named.
9. *Omnific-predicate questions.* Neither source knew the automorphisms report or the reconstruction section of the omnific Diophantine report; their language questions are re-scoped ([proposition 15.3](#) and [question 17.1](#)).
10. *Proof gap.* 04’s proof that a proper  $K_{\text{HOD}}$  is not intrinsically definable took the interval endpoints in the subfield ([proposition 5.4](#)).
11. *Bibliography.* 07 gave Gonshor’s book as Lecture Note Series 124; it is number 110, as 04 has it.
12. *Delivery files.* Both sources described their own archives (`.tex`, PDF, README). Those files are not shipped here; the shipped files are listed in the README.

### A.4 What each source inspected

**Source 04.** The repository was accessed through its GitHub connector at the pin. Inspection covered its README, the report catalogue, the introduction and provenance section of the birthday-cutoff/hereditary-set report, and the introduction and convention section of the omnific-Diophantine report. It was not a line-by-line audit of all reports or their preserved source manuscripts. No repository files were edited, no commit was made, and no Lean build was performed. The literature comparison used the supplied definable-real reference, primary works and author announcements on pointwise definability, HOD, surreal arithmetic, omnific normal forms, and quantifier elimination; the recent first-HOD-disagreement preprint was checked as

well. The cited works and the catalogue did not reveal the fixed-interval code or the support-subfield amplification. This limited negative search evidence cannot establish publication priority.

**Source 07.** The repository was inspected at the pin through its connected GitHub interface: the README, the catalogue, and the detailed guides for the universes, birthday and omnific Diophantine reports, whose cross-references identified the quotient report. These guides identify the statements, provenance and proof-review boundaries of the earlier manuscripts; they are not an exhaustive independent review of every proof or file. The literature search, on 23 September 2026, combined “definable surreal”, “ordinal definable surreal”, “surreal HOD”, “definable omnific”, “first undefinable ordinal surreal” and “definable surreal initial ordinal” with the named classical papers. The most relevant precedent located was Hamkins’s 2012 density answer; the epsilon-cutoff statement was checked against Corollary 5.5 of the original paper. No matching statement of the coding/core package was located; that supports describing these as proposed new formulations, and does not certify priority or exclude an equivalent result under different terminology.

**The merge.** Every cross-report label cited here was checked in the collection at the placement commit. The Bournez–Guilmant abstract was checked for the epsilon-number characterization.

## A.5 Files and checks

The source is self-contained apart from ordinary LaTeX packages: its bibliography is included below, and no external graphics or bibliography database is required. The two finite scripts shipped with the report are exact regression checks. Neither decides definability, approximates HOD, or replaces  $\omega$  by a large real number: that numerical substitution would destroy the exact non-Archimedean ordering relevant to the proofs.

**Source 04’s checks.** The script represents the nested exponents  $e_n$  by formal tags. Its recorded run passed 8,191 finite code round trips through sign length 12, 11 malformed-code rejections, 3,121 exact rational normalizations of the two-term map  $h$  with 3,120 strict order comparisons and 2 rejected endpoints, 2,157 formal infinitesimal floor cases, and 500 finite support-coset polynomial checks with 55,105 noncancelled formal support terms. A deliberately dependent exponent example correctly produced a vanishing polynomial, checking that the independence hypothesis is not silently omitted.

**Source 07’s checks.** The script checks finite normal-form bit encodings, their decoders and marker bounds, rational instances of the compression/inverse identities, and floor-correction examples. Its recorded run passed 8,191 bit codes (exhaustive for lengths 0 through 12), 4 malformed-code rejections, 555 rational compression/inverse pairs, 554 adjacent monotonicity comparisons, 1,161 floor-correction cases and 24 support-shift cases. It verifies no transfinite, set-theoretic satisfaction, HOD, epsilon-cutoff, elementarity, or core statement.

Both scripts were rerun for this merge on copies and reproduced their recorded results up to line endings.

## B Non-claims

Every limitation stated by either source is kept here; most also appear at the relevant statement.

## B.1 Source 04

- N4.1 The report is not refereed, not formalized in Lean, and makes no mapping to Lean declarations; no Lean build was performed.
- N4.2 The coding and transcendence results are proposed as original on the strength of a limited, non-exhaustive search; priority is not established, and no theorem settles a named open problem in the published literature.
- N4.3  $\mathbf{C}$  is not a ring homomorphism, is not definable in the pure ordered field, and has no arithmetic compatibility (the code of 0 is not 0).
- N4.4 “Ord-indexed independent” means only that finite subfamilies are independent. No transcendence basis, no Global Choice, no class sum, and no ordinary cardinal of a proper class is asserted; in a set model the assertions have their internal meaning.
- N4.5  $D_{\mathcal{O}}^V$  is not an internal set or class; there is no truth predicate for  $V$ .
- N4.6 For nonstandard models, codes are internal objects; the finite-ambiguity argument concerns genuinely finite sets.
- N4.7 The definable closure of  $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$  is not classified (see [proposition 15.3](#) for what the collection now proves).
- N4.8 Definability is not identified with computability or with the computable-surreal representations; normal-form uniqueness supplies no algorithm for infinite codes and no topological convergence of partial sums.
- N4.9  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  is not new; the contribution is the uniformity of the denominator; no priority is claimed for the general closure principle.
- N4.10 Termwise HOD membership is insufficient;  $K_{\text{HOD}}$  is not the Hahn field of HOD entries.
- N4.11 The widths in the localization are infinite monomial widths; a real width below 1 contains at most one omnific integer; no density is claimed.
- N4.12 The  $B_n$  escape asserts no  $m = n + 1$ , no closed form, no algorithm, and no cofinality of the  $d_n$ ; the sequence of the  $B_n$  is not uniformly definable.
- N4.13 The BCGHP corollary is a translation of a cited theorem, not a new surreal theorem;  $\lambda_H$  is not claimed to be regular.
- N4.14 Forcing: old definitions need not keep their truth or uniqueness;  $V = \text{HOD}$  is not preserved by arbitrary forcing; a generic filter over the actual universe is not assumed to be a set in it.
- N4.15 The surcomplex results use canonical coordinates; the abstract algebraically closed-field language behaves differently.
- N4.16 The questions are not published open problems and not omitted hypotheses.
- N4.17 The finite checks are regression tests only;  $\omega$  is never replaced by a real, and no infinite or set-theoretic theorem is checked.
- N4.18 The Chen–Hamkins–Yang announcement is not a black-box dependency; the set-coding mechanism is not new.
- N4.19 No nonclassical Diophantine, quotient, automorphism or bi-interpretation claim of the repository is a proof dependency.

- N4.20 The collector has a nondefinable coefficient only under the stated hypothesis that some member of  $\mathbb{R} \cap \text{HOD}$  is not parameter-free definable.
- N4.21 No birthday estimate for codes is claimed or used.
- N4.22 The existence of pointwise definable models is prior work (HLR).
- N4.23 No claim to have introduced HOD-valued surreal fields; the first-birthday threshold is the standard first subset disagreement.
- N4.24 The two-term code is an elementary observation, not a major novelty claim.

## B.2 Source 07

- N7.1 The main setting is an externally fixed transitive set model; its existence is not a ZFC theorem, and theorems quantified over it are conditional.
- N7.2 The core theorem and its corollaries are not claimed for nonstandard models.
- N7.3  $D_A^M$  is not claimed to be elementary in  $M$ ;  $K_A^M$  need not be represented by a set of  $M$ .
- N7.4 Provable uniqueness does not give absoluteness (the CH example).
- N7.5 There is no algorithm deciding definability; definability is not identified with computability.
- N7.6 The  $\mathcal{L}_{\text{exp}}$  results do not cover the restricted-analytic language or other uncountable languages.
- N7.7  $\mathbf{Q}$  is not onto an interval, is not ordered-field-definable, and uses  $\Omega$ , not  $\exp$ .
- N7.8 The birthday escape does not give density.
- N7.9 The set-coding lemma gives no canonical definable injection of sets into  $\mathbf{Oz}$ ; its use of Choice is existential.
- N7.10 Noncollection concerns sets of  $M$ , not classes.
- N7.11 The reverse gap is conditional; an internally uncountable ordinal need not have an undefinable predecessor.
- N7.12  $\rho$  is external, not a parameter (the Berry distinction).
- N7.13 An initial real closed subfield need not be the fraction field of its omnific part (real algebraic numbers with  $\mathbb{Z}$ ).
- N7.14 The finite-parameter discontinuity needs an externally uncountable ordinal.
- N7.15 Hamkins 2012, the old-field topology result, pointwise-model existence, birthday bi-interpretations, full-class denominator clearing and full-class finite quotients are credited, not claimed.
- N7.16 There is no exact definable closure for the real exponential field and no claim about particular transcendental reals.
- N7.17 The complexification depends on the chosen orientation and is not transferred to the pure algebraically closed-field language.
- N7.18 The full-class quotient theorem does not transfer to fragments; no Peano arithmetic and no stronger induction is inferred from finite-quotient agreement.
- N7.19 The questions are not published open problems.

- N7.20 The finite checks verify no transfinite, HOD, epsilon-cutoff, elementarity or core statement.
- N7.21 No Lean formalization; priority is not certified; novelty and correctness are separate, and many arguments are short combinations of established theorems.
- N7.22 Interdefinability by formula substitution need not preserve quantifier complexity; no optimal description-length theorem is asserted.
- N7.23 The Cohen example is relative to the existence of the ground model; the homogeneity argument does not assert that no new OD sets of arbitrary kind are added, nor that every forcing extension has the same HOD.
- N7.24 The  $\mathcal{L}_{\text{exp}}$  elementarity is proved formula by formula, with no truth predicate in  $M$ .
- N7.25 The self-similar value group is not a claim that  $K_A$  contains every Hahn series with entries in  $K_A$ .

### B.3 The merge

- NM.1 [Proposition 15.3](#) does not identify  $\text{dcl}_{\mathfrak{E}}(\emptyset)$ ; the omega-map expansion is not addressed.
- NM.2 `odg;q:size` is not answered; [remark 16.6](#) gives one instance.
- NM.3 Support-subfield amplification is not claimed for general same-ordinal inner models ([question 17.3](#)).
- NM.4 The agreement of HOD's internal omega map, exponential and normal forms with the ambient ones rests on the sources' recursion argument, not on the universes report, which covers arithmetic and cuts.

## C A small gallery of examples [07]

| Input or datum               | Constructed number                                              | What it illustrates                  |
|------------------------------|-----------------------------------------------------------------|--------------------------------------|
| $x = 0$                      | $Q(0) = \omega^{1/2}$                                           | A definable bounded omnific code     |
| $x = 3/4$                    | $Q(3/4) = \omega^{4/5}$                                         | Exact algebraic compression          |
| $x = -3/4$                   | $Q(-3/4) = \omega^{1/5}$                                        | Order preservation                   |
| $\alpha = 0, X = \emptyset$  | $E_0(X) = \omega^{3/4}$                                         | Marker and empty decoder             |
| $\alpha = 2, X = \{1\}$      | $\omega^{3/4} + \omega^{1/2} + 2\omega^{1/3}$                   | Finite bit assignment                |
| $x = \omega - \omega^{-1}$   | $\lfloor x \rfloor_{\mathbf{OZ}} = \omega - 1$                  | Negative infinitesimal correction    |
| $(a, b) = (\omega, 2\omega)$ | $Q_{a,b}(x) = \omega + \omega^{\tau(x)}$                        | Coding in an infinite-width interval |
| $s = 1$                      | $C(1) = \omega + 2\omega^{\omega^{-1}} + 3\omega^{\omega^{-2}}$ | The sign code (04)                   |

For the second row,  $\sqrt{1 + (3/4)^2} = 5/4$ , so  $\tau(3/4) = (1 + 3/5)/2 = 4/5$ . Every identity is an exact field identity, not a numerical approximation. The examples involving  $\omega$  are normal forms with symbolic exponents and are not evaluated by substituting a large real number for  $\omega$ .

## D Notation and reading guide

Table 3: Notation.

| Notation                             | Meaning                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------|
| $\mathbf{No}, \mathbf{Oz}, \Pi$      | Canonical surreal field, classical omnific-integer ring, purely infinite ideal (07's $\mathcal{J}$ ). |
| $b(s), \prec_s$                      | Ordinal length of the canonical sign sequence; proper prefix.                                         |
| $\omega^x = \Omega(x)$               | Conway monomial map; not $\exp(x \log \omega)$ .                                                      |
| $\underline{\alpha}$                 | The surreal all-plus ordinal $\alpha$ .                                                               |
| $\text{NF}(x), \text{supp}(x)$       | Whole indexed normal form; its exponent support.                                                      |
| $M, A$                               | A set model of ZFC (transitive where stated) and an external allowed parameter family.                |
| $D_A^M$                              | External definable closure of $A$ in $(M, \in)$ (07's $\text{Def}_M(A)$ ).                            |
| $K_A, I_A, O_A$                      | Ambiently $A$ -definable surreals, omnific integers, ordinals (07's $D_M(A), I_M(A), O_M(A)$ ).       |
| $\Pi_A, R_A$                         | $K_A \cap \Pi$ and $K_A \cap \mathbb{R}$ .                                                            |
| $K_{\text{HOD}}, I_{\text{HOD}}$     | Ordinal-definable surreals and omnific integers; $K_{\text{HOD}} = \mathbf{No}^{\text{HOD}}$ .        |
| $\mathcal{U}, \mathcal{U}_2$         | $\{z \in \mathbf{Oz} : \omega < z < \omega + \omega^{1/2}, \text{ct}(z) = 0\}$ and its two-term part. |
| $e_\alpha$                           | $\omega^{-(\alpha+1)}$ , a positive infinitesimal surreal.                                            |
| $\iota_\beta$                        | $1/(\underline{\beta} + 2)$ (07's $e_\beta$ ).                                                        |
| $\mathbf{C}, \mathbf{D}$             | 04's sign code and its partial inverse.                                                               |
| $\mathbf{T}, h$                      | 04's two-term code and its rational compression.                                                      |
| $\tau, \mathbf{Q}, \mathbf{Q}_{a,b}$ | 07's algebraic compression, monomial code and local code (07's $C, C_{a,b}$ ).                        |
| $\text{esc}(\alpha)$                 | 07's bounded code with birthday above $\alpha$ (07's $T$ ).                                           |
| $E_\alpha(X)$                        | 07's bit code of $X \subseteq \alpha$ .                                                               |
| $\text{Set}, X_a$                    | 04's decoder of sets from signs, and the ordinal code of $a$ (04's $E, B$ ).                          |
| $\zeta, q_*$                         | 04's collector map and collector.                                                                     |
| $\rho_A^M, \text{Core}_A^M$          | First ordinal of $M$ not $A$ -definable (external), and the maximal initial core (07's $K_M(A)$ ).    |
| $\epsilon, \varepsilon$              | An epsilon number; an infinitesimal.                                                                  |
| $\lambda_H$                          | First ordinal with a non-HOD subset; the first birthday of a non-OD surreal.                          |
| $B_n, d_n$                           | Fixed- $\Sigma_n$ uniquely definable surreals, and the definable ordinal surreal escaping that level. |
| $\mathbf{No}_{\leq \alpha}$          | Surreals of birthday at most $\alpha$ (07's $B_\alpha$ ).                                             |
| $\mathfrak{S}$                       | $(\mathbf{No}; +, -, \cdot, <, 0, 1, \mathbf{Oz})$ .                                                  |
| $\text{ct}, \text{st}, v$            | Constant term, finite standard part, normal-form valuation.                                           |

For the main new algebraic argument alone, read [sections 2, 5 and 8](#). For the definability concept and its model-theoretic limitations, read [sections 3, 7 and 12](#). For the initial core, read [sections 4 and 10](#). The codes in [section 6](#) connect the routes.

## References

- [1] J. H. Conway, *On Numbers and Games*, second edition, A K Peters, 2001; first edition, Academic Press, 1976.
- [2] H. Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986.
- [3] L. van den Dries and P. Ehrlich, Fields of surreal numbers and exponentiation, *Fundamenta Mathematicae* **167** (2001), 173–188. doi:10.4064/fm167-2-3. In particular Theorem 2.1, Lemmas 4.1–4.2, Proposition 4.7, and Corollary 5.5. To be read with [4].
- [4] L. van den Dries and P. Ehrlich, Erratum to “Fields of surreal numbers and exponentiation”, *Fundamenta Mathematicae* **168** (2001), 295–297. doi:10.4064/fm168-3-5.
- [5] O. Bournez and Q. Guilmant, Surreal fields stable under exponential and logarithmic functions, arXiv:2201.08199, 2022. <https://arxiv.org/abs/2201.08199>.
- [6] S. L’Innocente and V. Mantova, *A factorisation theory for generalised power series and omnific integers*, arXiv:1710.07304, version 5, January 22, 2024. <https://arxiv.org/abs/1710.07304>. Journal version: doi:10.1016/j.aim.2024.109513.
- [7] D. Marker, *Model Theory: An Introduction*, Graduate Texts in Mathematics 217, Springer, 2002. For the quantifier-elimination argument, see also the author’s *Introductory Lectures on Model Theory II: Quantifier Elimination*, October 14, 2020, <https://homepages.math.uic.edu/~marker/MSRI-2.pdf>.
- [8] L. van den Dries, *Tame Topology and o-Minimal Structures*, London Mathematical Society Lecture Note Series 248, Cambridge University Press, 1998.
- [9] P. Ehrlich and E. Kaplan, *Number systems with simplicity hierarchies: a generalization of Conway’s theory of surreal numbers II*, preprint, arXiv:1512.04001, version 1, 2015. <https://arxiv.org/abs/1512.04001v1>.
- [10] T. Jech, *Set Theory: The Third Millennium Edition, Revised and Expanded*, Springer Monographs in Mathematics, Springer, 2003. doi:10.1007/3-540-44761-X.
- [11] J. C. Shepherdson, A non-standard model for a free variable fragment of number theory, *Bulletin de l’Académie Polonaise des Sciences, Série des sciences mathématiques, astronomiques et physiques* **12** (1964), 79–86. The integer-part application to **Oz** is also stated explicitly in [6, Section 1.1].
- [12] J. D. Hamkins, D. Linetsky, and J. Reitz, Pointwise definable models of set theory, *Journal of Symbolic Logic* **78** (2013), no. 1, 139–156. <https://arxiv.org/abs/1105.4597>. Author’s discussion: <https://jdh.hamkins.org/pointwisedefinablemodelssettheory/>.
- [13] J. D. Hamkins, answer to “Definable map from all the ordinals to the surreal numbers with a dense image”, *MathOverflow*, 8 April 2012. <https://mathoverflow.net/questions/93468/definable-map-from-all-the-ordinals-to-the-surreal-numbers-with-a-dense-image>.
- [14] J. D. Hamkins and C. Leahy, Algebraicity and implicit definability in set theory, *Notre Dame Journal of Formal Logic* **57** (2016), no. 3, 431–439. <https://arxiv.org/abs/1305.5953>; doi:10.1215/00294527-3542326.

- [15] G. Goldberg and A. Poveda, Compactness phenomena in HOD, *Forum of Mathematics, Sigma* **13** (2025), e118. doi:10.1017/fms.2025.10070. Author's version: <https://math.berkeley.edu/~goldberg/Papers/CompactnessPhenomenaInHOD.pdf>.
- [16] T. Benhamou, J. Cummings, G. Goldberg, Y. Hayut, and A. Poveda, *Compactness phenomena in HOD and the Optimality of Magidor's Covering theorem*, arXiv:2608.24190, version 1, August 25, 2026. <https://arxiv.org/abs/2608.24190>. Theorems 1.1–1.2 supply the cited first-disagreement results; these are external inputs, not newly proved claims of this report.
- [17] J. D. Hamkins, *Surreal arithmetic is bi-interpretable with set theory*, CUNY Logic Workshop announcement, posted 4 February 2026 for a talk on 13 March 2026; joint work with Junhong Chen and Ruizhi Yang, described as work in progress. <https://jdh.hamkins.org/surreal-arithmetic-cuny-logic-workshop-march-2026/>. Cited as an announcement, not a refereed proof source.
- [18] Wikipedia contributors, *Definable real number*, user-specified starting reference, accessed September 23, 2026. [https://en.wikipedia.org/wiki/Definable\\_real\\_number](https://en.wikipedia.org/wiki/Definable_real_number). No theorem depends on this secondary reference.
- [19] V. Reshetnikov and contributors, *Surreal*, GitHub repository; the sources inspected it at commit 9a385d3957bdfe3d9ea79f9a524751c90bd2c894, 23 September 2026. <https://github.com/VladimirReshetnikov/Surreal>. The repository describes its manuscripts as AI-assisted research drafts and tracks formalization separately.
- [20] *Birthday Cutoffs and Hereditary Sets*, merged research report in [19], docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets/ (label prefix hset:).
- [21] *Surreal Fields Across Set-Theoretic Universes*, merged research report in [19], docs/foundations-and-computation/surreal-fields-across-universes/ (label prefix univ:).
- [22] *Omnific Integers and Omnific–Diophantine Geometry*, merged research report in [19], docs/surreal/omnific-diophantine-geometry/ (label prefix odg:).
- [23] *Set-Sized Quotients of Omnific Integers*, merged research report in [19], docs/surreal/set-sized-quotients-of-omnific-integers/ (label prefix osq:).
- [24] *Omnific-Preserving Automorphisms*, merged research report in [19], docs/surreal/omnific-preserving-automorphisms/ (label prefix opa:).